

S001 - Store Dashboard Screen Specification

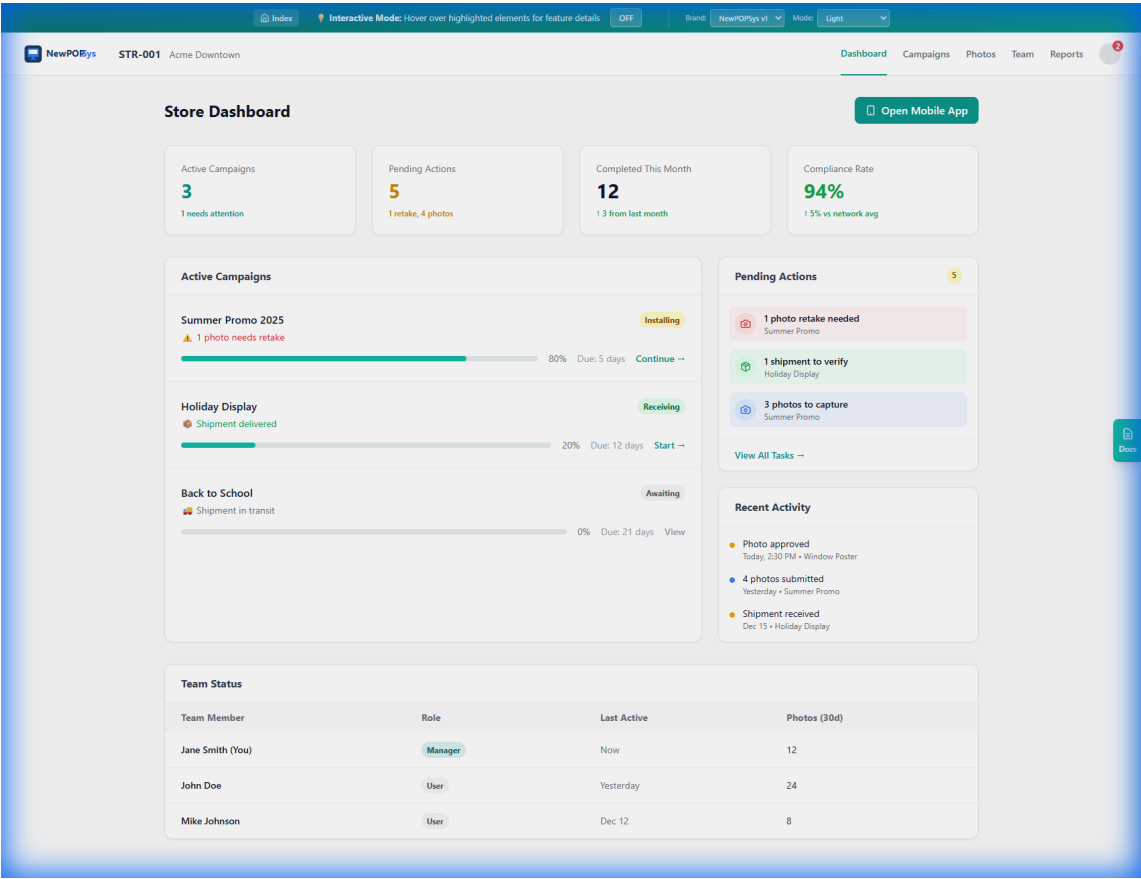
Module: StorePortal **Screen ID:** S001 **Route:** /store/dashboard **Version:** 1.0 **Last Updated:** 2026-01-01 **IEEE 830 Compliance:** Section 3.2 - Functional Requirements

1. Screen Overview

1.1 Purpose

The Store Dashboard serves as the primary landing page for store personnel, providing a consolidated view of campaign status, pending actions, and key performance metrics. It enables store managers and operators to quickly assess their workload and navigate to priority tasks.

1.2 Screenshot Reference



1.3 Source References

Document	Section
SUPP-001	Persona Workflows - Store Level
Screen Spec	SOW/06_Screen_Specs/S01_Dashboard.md

2. User Roles & Permissions

2.1 Authorized Roles

Role	Access Level	Persona ID
STORE_MANAGER	Full Access	P07
STORE_OPERATOR	Limited Access	P08

2.2 Role-Based Display Rules

Feature	STORE_MANAGER	STORE_OPERATOR
View all store campaigns	Yes	Yes (assigned only)
View team activity section	Yes	No
Access team management link	Yes	No
View store reports link	Yes	No
View compliance rate KPI	Yes	No
Execute campaign tasks	Yes	Yes

2.3 Permission Requirements

Requirement ID	Description	Priority
REQ-S001-SEC-001	User must have active membership with store to access dashboard	Must
REQ-S001-SEC-002	Dashboard data scoped to user's assigned store only	Must
REQ-S001-SEC-003	STORE_OPERATOR sees only their own pending actions	Must
REQ-S001-SEC-004	STORE_MANAGER sees all team members' pending actions	Must

3. UI Components

3.1 Component Inventory

Component ID	Component Name	Type	Description
S001-C001	Page Header	Header	"Dashboard" title with store name
S001-C002	KPI Cards Row	Card Grid	4 key performance indicator cards
S001-C003	Active Campaigns List	Data List	Expandable campaign entries
S001-C004	Pending Actions Panel	Action List	Priority-sorted task list

S001-C005	Recent Activity Feed	Timeline	Chronological activity stream
S001-C006	Team Status Card	Summary Card	Team member overview (Manager only)
S001-C007	Quick Actions Bar	Button Group	Primary navigation shortcuts

3.2 KPI Cards Specification

Card ID	Metric	Calculation	Visual
KPI-001	Active Campaigns	COUNT(assignments WHERE status NOT IN ['COMPLETE', 'WAIVED'])	Number badge
KPI-002	Pending Actions	COUNT(actions WHERE status = 'PENDING')	Number with urgency color
KPI-003	Completed This Month	COUNT(assignments WHERE completed_at >= first_of_month)	Number with trend arrow
KPI-004	Compliance Rate	(on_time_completions / total_completions) * 100	Percentage gauge

3.3 Layout Structure

Dashboard		Store: #12345	
[KPI-001]	[KPI-002]	[KPI-003]	[KPI-004]
Active	Pending	Completed	Compliance
3	5	12	94%
Active Campaigns		Pending Actions	
Campaign 1	[>]	⚠ Receive shipment	
Campaign 2	[>]	○ Upload photos	
Campaign 3	[>]	○ Complete survey	
Recent Activity		Team Status (Manager only)	
Jane uploaded photo		4 active members	
John completed survey		2 pending invitations	

3.4 Component Requirements

Requirement ID	Description	Priority
REQ-S001-UI-001	KPI cards must display loading skeleton during data fetch	Must

REQ-S001-UI-002	Active campaigns list supports expand/collapse interaction	Must
REQ-S001-UI-003	Pending actions sorted by urgency (due date ascending)	Must
REQ-S001-UI-004	Recent activity shows last 10 events with relative timestamps	Should
REQ-S001-UI-005	Team status card hidden for STORE_OPERATOR role	Must
REQ-S001-UI-006	Dashboard auto-refreshes every 5 minutes	Should

4. Data Requirements

4.1 Entity Dependencies

Entity	Table Name	Fields Required	Relationship
Store	stores	id, name, external_store_guid, status	Primary
StoreAssignment	store_assignments	id, campaign_id, store_id, status, pinned_layout_id	Many-to-One
Campaign	campaigns	id, name, install_start, install_end, status	Via Assignment
PhotoUpload	photo_uploads	id, assignment_item_id, review_status, created_at	Via Assignment
IssueRequest	issue_requests	id, store_assignment_id, status, created_at	Via Assignment
User	users	id, name, email, is_active	Via Membership
Membership	memberships	user_id, store_id, role	Join Table

4.2 Data Query Specification

```
-- Dashboard aggregate query
SELECT
  s.id as store_id,
  s.name as store_name,
  COUNT(DISTINCT CASE WHEN sa.status NOT IN ('COMPLETE', 'WAIVED') THEN sa.id END) as
active_campaigns,
  COUNT(DISTINCT CASE WHEN sa.status = 'COMPLETE'
    AND sa.completed_at >= DATE_TRUNC('month', NOW()) THEN sa.id END) as
completed_this_month,
  (SELECT COUNT(*) FROM memberships m
   JOIN users u ON m.user_id = u.id
   WHERE m.store_id = s.id AND u.is_active = true) as active_team_members
FROM stores s
LEFT JOIN store_assignments sa ON sa.store_id = s.id AND sa.deleted_at IS NULL
WHERE s.id = :storeId AND s.deleted_at IS NULL
GROUP BY s.id, s.name
```

4.3 Data Requirements

Requirement ID	Description	Priority
REQ-S001-DATA-001	All queries must filter by tenant_id from JWT	Must
REQ-S001-DATA-002	Soft-deleted records excluded from all counts	Must
REQ-S001-DATA-003	Campaign dates displayed in store's local timezone	Must
REQ-S001-DATA-004	Activity feed limited to last 30 days	Should

5. Business Rules & Validation

5.1 Display Rules

Rule ID	Rule Description	Condition	Action
BR-S001-001	Urgent action highlighting	Due date <= today	Display red badge
BR-S001-002	Campaign phase badge	Based on StorePhase enum	Show appropriate status color
BR-S001-003	Compliance rate threshold	Rate < 75%	Display warning indicator
BR-S001-004	Empty state handling	No active campaigns	Show "No active campaigns" message

5.2 StorePhase Display Mapping

StorePhase	Display Label	Badge Color
AWAITING_SHIPMENT	Awaiting Shipment	Gray
SHIPMENT_IN_TRANSIT	In Transit	Blue
READY_TO_RECEIVE	Ready to Receive	Yellow
RECEIVING	Receiving	Yellow
READY_TO_INSTALL	Ready to Install	Orange
INSTALLING	Installing	Orange
PENDING_REVIEW	Pending Review	Purple
COMPLETE	Complete	Green
NEEDS_ATTENTION	Needs Attention	Red
WAIVED	Waived	Gray

5.3 Business Rule Requirements

Requirement ID	Description	Priority
REQ-S001-BR-001	Campaigns sorted by install_end date (earliest first)	Must
REQ-S001-BR-002	NEEDS_ATTENTION status triggers visual alert	Must
REQ-S001-BR-003	Pending actions derive from incomplete workflow steps	Must
REQ-S001-BR-004	Completed campaigns remain visible for 30 days post-completion	Should

6. API Integration Points

6.1 API Endpoints

Endpoint	Method	Purpose	Response
/api/stores/{storeId}/dashboard	GET	Fetch dashboard aggregate data	DashboardDTO
/api/stores/{storeId}/assignments	GET	List active assignments	Assignment[]
/api/stores/{storeId}/activity	GET	Recent activity feed	ActivityEvent[]

6.2 Request/Response Specifications

GET /api/stores/{storeId}/dashboard

Request Headers:

Authorization: Bearer {jwt_token}
X-Tenant-ID: {tenant_uuid}

Response Schema:

```
{
  "store": {
    "id": "uuid",
    "name": "string",
    "externalGuid": "string"
  },
  "metrics": {
    "activeCampaigns": "number",
    "pendingActions": "number",
    "completedThisMonth": "number",
    "complianceRate": "number"
  },
  "campaigns": [
    {
      "id": "uuid",
      "name": "string",
      "phase": "StorePhase",
      "installStart": "ISO8601",
      "installEnd": "ISO8601",
      "pendingTasks": ["string"]
    }
  ]
}
```

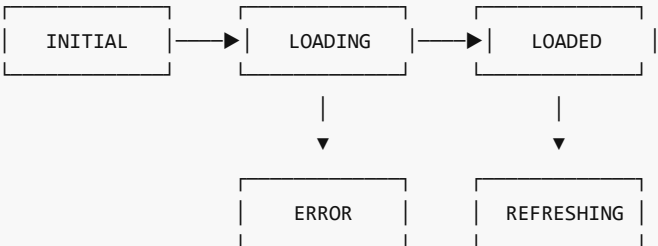
```
    }
  ],
  "recentActivity": [
    {
      "id": "uuid",
      "type": "string",
      "actor": "string",
      "description": "string",
      "timestamp": "ISO8601"
    }
  ],
  "teamStatus": {
    "activeMembers": "number",
    "pendingInvitations": "number"
  }
}
```

6.3 API Requirements

Requirement ID	Description	Priority
REQ-S001-API-001	Dashboard endpoint returns within 500ms	Must
REQ-S001-API-002	Response includes cache headers (max-age: 60)	Should
REQ-S001-API-003	404 returned if store not found or user lacks access	Must
REQ-S001-API-004	teamStatus omitted from response for STORE_OPERATOR	Must

7. State Transitions

7.1 Dashboard Load States



7.2 State Descriptions

State	Description	UI Behavior
INITIAL	Component mounted	Show loading skeleton
LOADING	API request in flight	Display skeleton, disable interactions
LOADED	Data successfully retrieved	Render dashboard components

ERROR	API request failed	Show error message with retry button
REFRESHING	Background data refresh	Subtle loading indicator, data remains visible

7.3 State Requirements

Requirement ID	Description	Priority
REQ-S001-STATE-001	Loading state displays within 100ms of navigation	Must
REQ-S001-STATE-002	Error state provides actionable retry option	Must
REQ-S001-STATE-003	Refreshing state does not interrupt user interaction	Must

8. Error Handling

8.1 Error Scenarios

Error Code	Scenario	User Message	Recovery Action
401	Unauthorized/expired token	"Session expired. Please log in again."	Redirect to login
403	No store access	"You don't have access to this store."	Redirect to store selector
404	Store not found	"Store not found."	Redirect to store selector
500	Server error	"Unable to load dashboard. Please try again."	Show retry button
NETWORK	Connection failed	"Connection lost. Check your internet."	Show retry with offline indicator

8.2 Partial Load Handling

Component	Failure Behavior
KPI Cards	Show "---" placeholder, log error
Campaign List	Show empty state with error message
Activity Feed	Show "Unable to load activity"
Team Status	Hide component, log error

8.3 Error Requirements

Requirement ID	Description	Priority
REQ-S001-ERR-001	All API errors logged with correlation ID	Must
REQ-S001-ERR-002	User-facing errors do not expose technical details	Must

REQ-S001-ERR-003	Retry attempts use exponential backoff	Should
REQ-S001-ERR-004	Partial failures do not block entire dashboard	Should

9. Accessibility Requirements

9.1 WCAG 2.1 AA Compliance

Requirement ID	WCAG Criterion	Implementation
REQ-S001-A11Y-001	1.1.1 Non-text Content	Alt text for all status icons and badges
REQ-S001-A11Y-002	1.3.1 Info and Relationships	Semantic HTML structure with proper headings
REQ-S001-A11Y-003	1.4.3 Contrast	Minimum 4.5:1 contrast for all text
REQ-S001-A11Y-004	2.1.1 Keyboard	All interactive elements keyboard accessible
REQ-S001-A11Y-005	2.4.4 Link Purpose	Descriptive link text for all navigation
REQ-S001-A11Y-006	4.1.2 Name, Role, Value	ARIA labels for dynamic content regions

9.2 Screen Reader Support

Component	ARIA Implementation
KPI Cards	role="status", aria-live="polite" for updates
Campaign List	role="list", aria-expanded for expandable items
Pending Actions	role="list", aria-describedby for urgency
Activity Feed	role="feed", aria-busy during refresh

9.3 Keyboard Navigation

Key	Action
Tab	Move between interactive elements
Enter/Space	Activate focused element
Arrow Down	Expand campaign details
Arrow Up	Collapse campaign details
Escape	Close any open panels

10. Acceptance Criteria

10.1 Functional Requirements

Requirement ID	Description	Test Criteria	Priority
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REQ-S001-FR-001	Dashboard displays 4 KPI cards	All 4 metrics visible with correct values	Must
REQ-S001-FR-002	Active campaigns list shows current assignments	List matches database query results	Must
REQ-S001-FR-003	Pending actions sorted by urgency	Earliest due date appears first	Must
REQ-S001-FR-004	Campaign cards expand to show details	Click expands, shows tasks and dates	Must
REQ-S001-FR-005	Recent activity shows last 10 events	Events display with relative timestamps	Should
REQ-S001-FR-006	Team status visible to STORE_MANAGER only	Component hidden for STORE_OPERATOR	Must
REQ-S001-FR-007	Dashboard auto-refreshes every 5 minutes	Data updates without page reload	Should
REQ-S001-FR-008	Quick actions navigate to correct screens	Each action links to proper route	Must
REQ-S001-FR-009	Campaign click navigates to campaign detail	Route: /store/campaigns/{id}	Must
REQ-S001-FR-010	Empty state displayed when no campaigns	Message shown when list is empty	Must

10.2 Non-Functional Requirements

Requirement ID	Description	Metric	Priority
REQ-S001-NFR-001	Page load time	< 2 seconds on 3G connection	Must
REQ-S001-NFR-002	API response time	< 500ms for dashboard endpoint	Must
REQ-S001-NFR-003	Mobile responsiveness	Renders correctly on 320px viewport	Must
REQ-S001-NFR-004	Offline indicator	Displays when connection lost	Should

10.3 Traceability Matrix

Requirement	Source Document	Section
REQ-S001-FR-001	S01_Dashboard.md	UI Components
REQ-S001-FR-002	S01_Dashboard.md	Active Campaigns
REQ-S001-SEC-001	SUPP-003	Store Level Permissions
REQ-S001-DATA-001	3.1_Database_Model.md	Multi-Tenancy Model

Document History

Version	Date	Author	Changes
1.0	2026-01-01	System	Initial SRS specification

Document Status: Complete IEEE 830 Compliance: Section 3.2 - Functional Requirements

S002 - Campaign History Screen Specification

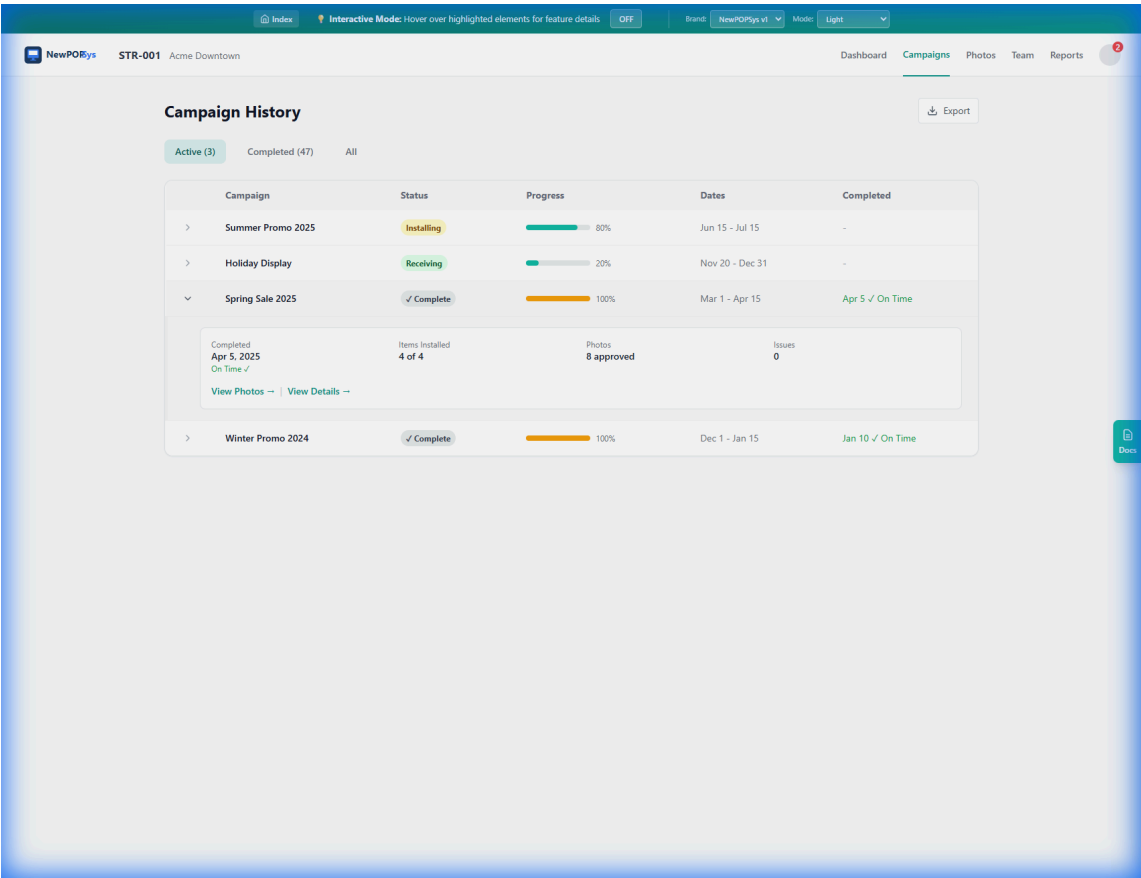
Module: StorePortal **Screen ID:** S002 **Route:** /store/campaigns **Version:** 1.0 **Last Updated:** 2026-01-01 **IEEE 830 Compliance:** Section 3.2 - Functional Requirements

1. Screen Overview

1.1 Purpose

The Campaign History screen provides store personnel with a comprehensive view of all campaign assignments, both active and completed. Users can filter by status, view detailed campaign information, track execution progress, and access historical campaign data for compliance and audit purposes.

1.2 Screenshot Reference



1.3 Source References

Document	Section
SUPP-001	Persona Workflows - Store Level
Screen Spec	SOW/06_Screen_Specs/S02_Campaign_History.md
Database Model	Section 3.1 - Campaign & Assignment Tables

2. User Roles & Permissions

2.1 Authorized Roles

Role	Access Level	Persona ID
STORE_MANAGER	Full Access	P07
STORE_OPERATOR	View + Execute	P08

2.2 Role-Based Capabilities

Capability	STORE_MANAGER	STORE_OPERATOR
View all store campaigns	Yes	Yes (assigned only)
View campaign details	Yes	Yes
Export campaign list	Yes	No
View all team submissions	Yes	Own submissions only
Access campaign analytics	Yes	No

2.3 Permission Requirements

Requirement ID	Description	Priority
REQ-S002-SEC-001	User must have active store membership	Must
REQ-S002-SEC-002	Campaign list scoped to user's assigned store	Must
REQ-S002-SEC-003	STORE_OPERATOR limited to assigned campaigns only	Should
REQ-S002-SEC-004	Export functionality restricted to STORE_MANAGER	Must

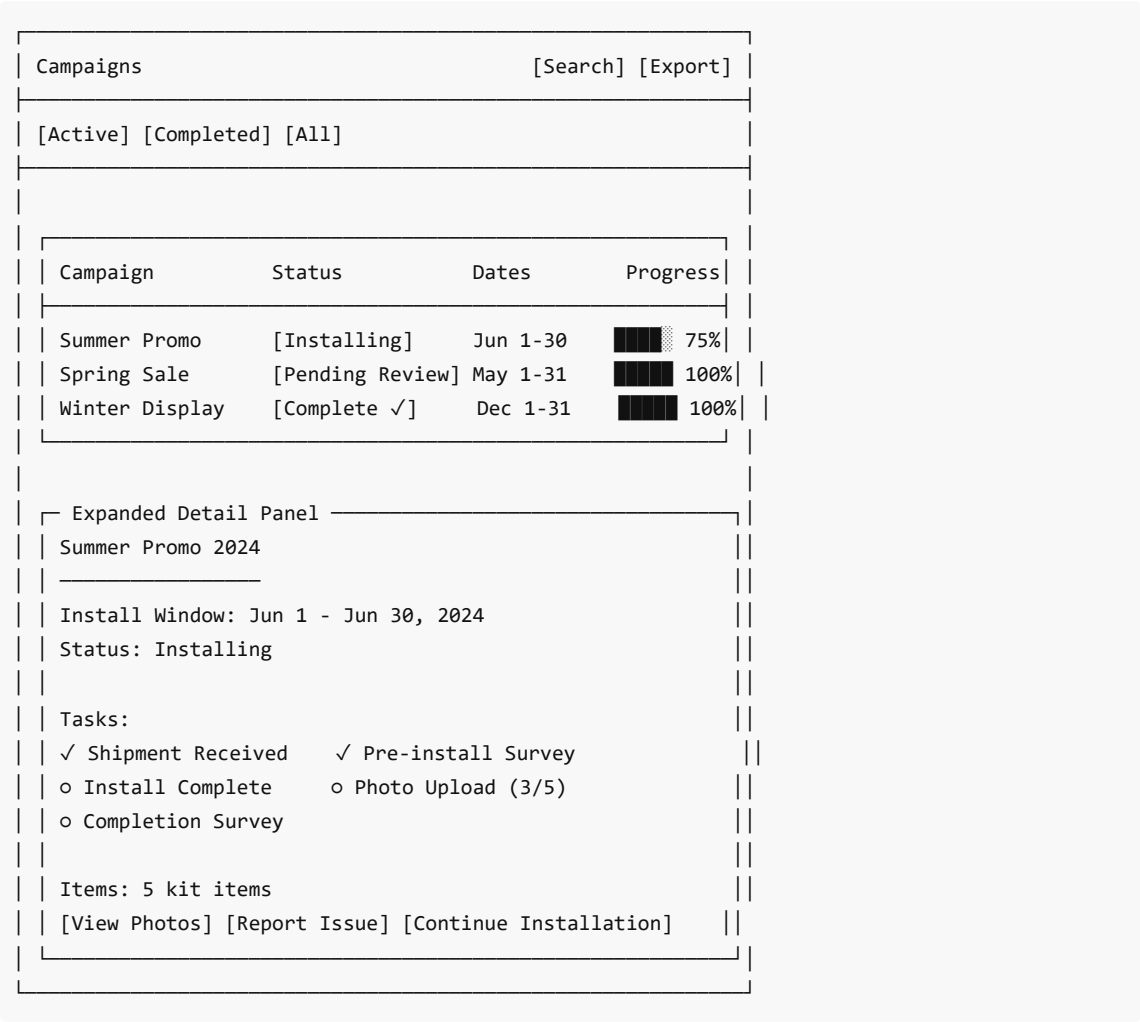
3. UI Components

3.1 Component Inventory

Component ID	Component Name	Type	Description
S002-C001	Page Header	Header	"Campaigns" with filter controls

S002-C002	Status Tabs	Tab Bar	Active, Completed, All filters
S002-C003	Campaign List	Data Table	Sortable campaign entries
S002-C004	Search Bar	Input	Campaign name/ID search
S002-C005	Detail Panel	Slide Panel	Expandable campaign details
S002-C006	Phase Badge	Status Badge	StorePhase indicator
S002-C007	Progress Bar	Progress	Visual completion indicator
S002-C008	Export Button	Button	CSV download trigger
S002-C009	Empty State	Message	No campaigns message

3.2 Layout Structure



3.3 Component Requirements

Requirement ID	Description	Priority
REQ-S002-UI-001	Campaign list supports column sorting	Must

REQ-S002-UI-002	Status tabs filter list dynamically without page reload	Must
REQ-S002-UI-003	Detail panel slides in from right on row click	Must
REQ-S002-UI-004	Progress bar reflects actual task completion percentage	Must
REQ-S002-UI-005	Search filters by campaign name or external ID	Should
REQ-S002-UI-006	List displays 20 items per page with pagination	Must

4. Data Requirements

4.1 Entity Dependencies

Entity	Table Name	Fields Required	Relationship
StoreAssignment	store_assignments	id, campaign_id, store_id, status, pinned_layout_id, completed_at	Primary
Campaign	campaigns	id, name, code, install_start, install_end, status	Many-to-One
AssignmentItem	assignment_items	id, assignment_id, kit_item_id, slot_id	One-to-Many
KitItem	kit_items	id, name, sku, quantity	Via AssignmentItem
PhotoUpload	photo_uploads	id, assignment_item_id, review_status	Via AssignmentItem
ReceiveVerification	receive_verifications	id, assignment_id, verified_at	One-to-One
CompletionAttestation	completion_attestations	id, assignment_id, attested_by	One-to-One

4.2 Data Query Specification

```
-- Campaign list query
SELECT
  sa.id as assignment_id,
  sa.status as phase,
  sa.completed_at,
  c.id as campaign_id,
  c.name as campaign_name,
  c.code as campaign_code,
  c.install_start,
  c.install_end,
  COUNT(DISTINCT ai.id) as total_items,
  COUNT(DISTINCT CASE WHEN pu.review_status = 'APPROVED' THEN pu.id END) as approved_photos,
  CASE WHEN rv.id IS NOT NULL THEN true ELSE false END as received,
  CASE WHEN ca.id IS NOT NULL THEN true ELSE false END as completed
FROM store_assignments sa
```

```
JOIN campaigns c ON sa.campaign_id = c.id
LEFT JOIN assignment_items ai ON ai.store_assignment_id = sa.id
LEFT JOIN photo_uploads pu ON pu.assignment_item_id = ai.id
LEFT JOIN receive_verifications rv ON rv.store_assignment_id = sa.id
LEFT JOIN completion_attestations ca ON ca.store_assignment_id = sa.id
WHERE sa.store_id = :storeId
      AND sa.deleted_at IS NULL
      AND c.deleted_at IS NULL
GROUP BY sa.id, c.id, rv.id, ca.id
ORDER BY c.install_end DESC
```

4.3 Data Requirements

Requirement ID	Description	Priority
REQ-S002-DATA-001	Campaigns ordered by install_end date descending	Must
REQ-S002-DATA-002	Include item counts and photo statistics per assignment	Must
REQ-S002-DATA-003	Soft-deleted records excluded from all queries	Must
REQ-S002-DATA-004	Completed campaigns retained for 2 years	Should

5. Business Rules & Validation

5.1 Status Tab Filtering

Tab	Filter Condition	StorePhase Values Included
Active	status NOT IN ('COMPLETE', 'WAIVED')	AWAITING_SHIPMENT, SHIPMENT_IN_TRANSIT, READY_TO_RECEIVE, RECEIVING, READY_TO_INSTALL, INSTALLING, PENDING_REVIEW, NEEDS_ATTENTION, REOPENED
Completed	status IN ('COMPLETE', 'WAIVED')	COMPLETE, WAIVED
All	No filter	All values

5.2 StorePhase Status Display

StorePhase	Display Label	Badge Color	Icon
AWAITING_SHIPMENT	Awaiting Shipment	Gray	clock
SHIPMENT_IN_TRANSIT	In Transit	Blue	truck
READY_TO_RECEIVE	Ready to Receive	Yellow	package
RECEIVING	Receiving	Yellow	package-open
READY_TO_INSTALL	Ready to Install	Orange	tools

INSTALLING	Installing	Orange	hammer
PENDING_REVIEW	Pending Review	Purple	eye
COMPLETE	Complete	Green	check-circle
NEEDS_ATTENTION	Needs Attention	Red	alert-triangle
WAIVED	Waived	Gray	skip-forward
REOPENED	Reopened	Orange	refresh

5.3 Progress Calculation

Component	Weight	Calculation
Shipment Received	10%	receive_verification EXISTS
Pre-install Survey	10%	survey_response EXISTS WHERE type='PRE_INSTALL'
All Items Installed	50%	all assignment_items have status='INSTALLED'
All Photos Approved	20%	all required photos have review_status='APPROVED'
Completion Survey	10%	completion_attestation EXISTS

5.4 Business Rule Requirements

Requirement ID	Description	Priority
REQ-S002-BR-001	NEEDS_ATTENTION campaigns highlighted with alert icon	Must
REQ-S002-BR-002	Overdue campaigns (past install_end) show warning indicator	Must
REQ-S002-BR-003	Progress percentage rounds to nearest whole number	Should
REQ-S002-BR-004	WAIVED campaigns show reason in detail panel	Should

6. API Integration Points

6.1 API Endpoints

Endpoint	Method	Purpose	Response
/api/stores/{storeId}/assignments	GET	List campaign assignments	Assignment[]
/api/assignments/{id}	GET	Get assignment details	AssignmentDetail
/api/stores/{storeId}/assignments/export	GET	Export campaign list as CSV	File download

6.2 Request/Response Specifications

GET /api/stores/{storeId}/assignments

Query Parameters:

Parameter	Type	Required	Description
status	string	No	Filter: active, completed, all
search	string	No	Search by name/code
page	number	No	Page number (default: 1)
limit	number	No	Items per page (default: 20)

Response Schema:

```
{
  "data": [
    {
      "id": "uuid",
      "campaignId": "uuid",
      "campaignName": "string",
      "campaignCode": "string",
      "phase": "StorePhase",
      "installStart": "ISO8601",
      "installEnd": "ISO8601",
      "progress": "number",
      "itemCount": "number",
      "photoStats": {
        "required": "number",
        "uploaded": "number",
        "approved": "number"
      },
      "completedAt": "ISO8601 | null"
    }
  ],
  "pagination": {
    "page": "number",
    "limit": "number",
    "total": "number",
    "pages": "number"
  }
}
```

GET /api/assignments/{id}

Query Parameters:

Parameter	Type	Required	Description
include	string	No	Comma-separated: items,photos,tasks

Response Schema:

```
{
  "id": "uuid",
```

```

"campaign": {
  "id": "uuid",
  "name": "string",
  "code": "string",
  "installStart": "ISO8601",
  "installEnd": "ISO8601"
},
"phase": "StorePhase",
"progress": "number",
"tasks": [
  {
    "name": "string",
    "status": "pending | complete",
    "completedAt": "ISO8601 | null"
  }
],
"items": [
  {
    "id": "uuid",
    "name": "string",
    "sku": "string",
    "quantity": "number",
    "slotCode": "string",
    "photoCount": "number",
    "photoApproved": "boolean"
  }
],
"receiveVerification": {
  "verifiedAt": "ISO8601",
  "verifiedBy": "string"
},
"completionAttestation": {
  "attestedAt": "ISO8601",
  "attestedBy": "string"
}
}

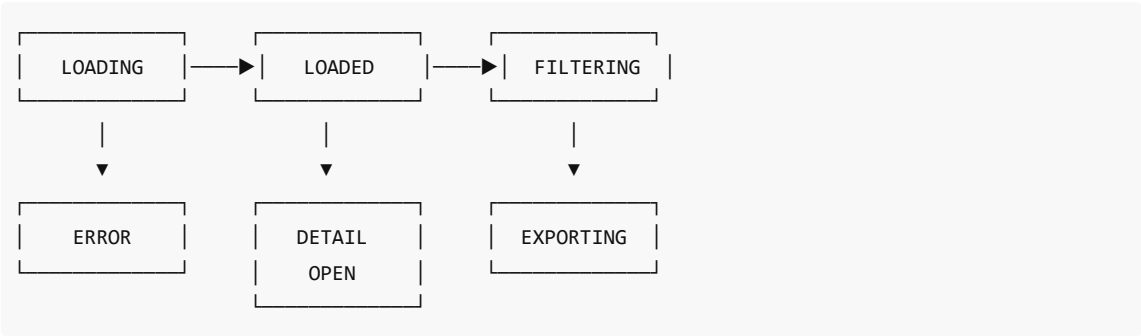
```

6.3 API Requirements

Requirement ID	Description	Priority
REQ-S002-API-001	List endpoint returns within 500ms	Must
REQ-S002-API-002	Detail endpoint returns within 300ms	Must
REQ-S002-API-003	Export generates CSV within 10 seconds	Should
REQ-S002-API-004	Pagination required for lists > 20 items	Must

7. State Transitions

7.1 Page States



7.2 State Descriptions

State	Description	UI Behavior
LOADING	Initial data fetch	Show skeleton loader
LOADED	Data displayed	Full list visible
FILTERING	Tab or search change	Update list, maintain position
DETAIL_OPEN	Row expanded	Slide panel visible
EXPORTING	CSV generation	Show progress, disable export button
ERROR	API failure	Show error message, retry option

7.3 State Requirements

Requirement ID	Description	Priority
REQ-S002-STATE-001	Tab changes update URL query parameter	Should
REQ-S002-STATE-002	Detail panel state preserved on tab change	Should
REQ-S002-STATE-003	Export state shows progress indicator	Must

8. Error Handling

8.1 Error Scenarios

Error Code	Scenario	User Message	Recovery Action
401	Unauthorized	"Session expired"	Redirect to login
403	No access	"You don't have access to this store"	Redirect to store selector
404	Campaign not found	"Campaign not found"	Remove from list, show toast
500	Server error	"Unable to load campaigns"	Retry button

EXPORT_FAIL	Export generation failed	"Export failed. Please try again."	Retry button
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8.2 Validation Errors

Scenario	Handling
Invalid date range	Show inline error, reset to defaults
Search no results	Show "No campaigns found" empty state
Page out of bounds	Redirect to last valid page

8.3 Error Requirements

Requirement ID	Description	Priority
REQ-S002-ERR-001	Failed export shows error toast	Must
REQ-S002-ERR-002	Network errors display retry option	Must
REQ-S002-ERR-003	Invalid campaign removed from list gracefully	Should

9. Accessibility Requirements

9.1 WCAG 2.1 AA Compliance

Requirement ID	WCAG Criterion	Implementation
REQ-S002-A11Y-001	1.3.1 Info and Relationships	Table uses proper th/td structure
REQ-S002-A11Y-002	1.4.1 Use of Color	Status not conveyed by color alone
REQ-S002-A11Y-003	2.1.1 Keyboard	Tab navigation through all controls
REQ-S002-A11Y-004	2.4.3 Focus Order	Logical focus order in detail panel
REQ-S002-A11Y-005	4.1.1 Parsing	Valid HTML, no duplicate IDs
REQ-S002-A11Y-006	4.1.2 Name, Role, Value	ARIA labels for status badges

9.2 Screen Reader Support

Component	ARIA Implementation
Campaign Table	role="grid" with sortable column headers
Status Badge	aria-label="Status: {phase}"
Progress Bar	role="progressbar", aria-valuenow, aria-valuemin, aria-valuemax
Detail Panel	role="dialog", aria-modal="true"
Tabs	role="tablist" with aria-selected

9.3 Keyboard Navigation

Key	Action
Tab	Move between interactive elements
Enter	Open campaign detail / activate button
Escape	Close detail panel
Arrow Left/Right	Switch between status tabs
Arrow Up/Down	Navigate table rows

10. Acceptance Criteria

10.1 Functional Requirements

Requirement ID	Description	Test Criteria	Priority
REQ-S002-FR-001	Campaign list displays all store assignments	List matches database records	Must
REQ-S002-FR-002	Active tab shows only in-progress campaigns	Filter excludes COMPLETE/WAIVED	Must
REQ-S002-FR-003	Completed tab shows finished campaigns	Filter includes only COMPLETE/WAIVED	Must
REQ-S002-FR-004	Search filters by campaign name	Real-time filtering as user types	Must
REQ-S002-FR-005	Row click opens detail panel	Panel slides in from right	Must
REQ-S002-FR-006	Detail panel shows all tasks	Checklist with completion status	Must
REQ-S002-FR-007	Progress bar reflects actual completion	Percentage matches calculation	Must
REQ-S002-FR-008	Export generates CSV file	File downloads with correct data	Must
REQ-S002-FR-009	Status badges use correct colors	Colors match StorePhase mapping	Must
REQ-S002-FR-010	Pagination works for large lists	Previous/next navigate correctly	Must
REQ-S002-FR-011	Overdue campaigns show warning	Visual indicator for past install_end	Should
REQ-S002-FR-012	Empty state shown when no campaigns	Message displayed for empty list	Must

10.2 Non-Functional Requirements

Requirement ID	Description	Metric	Priority
REQ-S002-NFR-001	Page load time	< 2 seconds	Must
REQ-S002-NFR-002	Search response time	< 300ms	Must
REQ-S002-NFR-003	Export completion	< 10 seconds for 1000 campaigns	Should
REQ-S002-NFR-004	Mobile responsive	Works on 320px viewport	Must

10.3 Traceability Matrix

Requirement	Source Document	Section
REQ-S002-FR-001	S02_Campaign_History.md	Data Model Map
REQ-S002-FR-005	S02_Campaign_History.md	Expanded Detail Panel
REQ-S002-SEC-001	SUPP-003	Store Level Permissions
REQ-S002-DATA-001	3.1_Database_Model.md	Store Assignments

Document History

Version	Date	Author	Changes
1.0	2026-01-01	System	Initial SRS specification

Document Status: Complete IEEE 830 Compliance: Section 3.2 - Functional Requirements

S003 Photo Gallery - Screen Specification

SRS Section: 5.9.3 | **Module:** Store Portal | **Version:** 1.0 **IEEE 830 Reference:** Section 3.2 - Functional Requirements
Source Documents:

- [S03 Photo Gallery Screen Spec](#)
- [SUPP-018 Photo Review](#)
- [SUPP-037 Store Surveys](#) **Last Updated:** 2026-01-01

1. Screen Overview

1.1 Purpose

The Photo Gallery screen provides store personnel with a centralized view of all installation proof photos submitted for their store. It enables browsing, filtering, and reviewing photo upload history across campaigns, with detailed status visibility for approved, pending, rejected, and superseded photos.

1.2 Route Configuration

Attribute	Value
Route Path	/store/photos
Route Type	Protected (Authentication Required)
Lazy Loading	Yes
Mobile Support	Full Responsive

1.3 Screen Context

Attribute	Description
Primary Purpose	Browse and review all store photo submissions
Entry Points	Store Dashboard quick link, Campaign History photo link
Exit Points	Dashboard, Campaign History, Photo Capture (mobile)
Session Scope	Store context from authenticated membership

1.4 Screenshot Reference

 Photo Gallery Screen

2. User Roles & Permissions

2.1 Authorized Roles

Role	Access Level	Restrictions
STORE_MANAGER (P07)	Full Access	Own store photos only
STORE_OPERATOR (P08)	Read Access	Own store photos only

2.2 Permission Requirements

Permission	STORE_MANAGER	STORE_OPERATOR
View photo gallery	Y	Y
View photo details/lightbox	Y	Y
Download individual photos	Y	Y
Download bulk photos	Y	N
View rejection reasons	Y	Y
View admin comments	Y	Y
Filter by team member	Y	N

2.3 Data Scoping Rules

Rule ID	Description
REQ-S003-SEC-001	Photos filtered to authenticated user's store via store_assignments.store_id
REQ-S003-SEC-002	Store membership validated from JWT token claims
REQ-S003-SEC-003	Cross-store photo access blocked at API and database layers

3. UI Components

3.1 Component Hierarchy

```
PhotoGalleryPage
├─ PageHeader
│   ├── TitleSection ("Photo Gallery")
│   └─ StatusSummary (count by status)
├─ FilterBar
│   ├── CampaignFilter (dropdown)
│   ├── StatusFilter (dropdown)
│   ├── DateRangeFilter (dropdown)
│   ├── ItemTypeFilter (multi-select)
│   └─ UploadedByFilter (dropdown) [Store Manager only]
├─ ViewToggle
│   ├── GridViewButton
│   └─ ListViewButton
├─ ActionBar
│   └─ BulkDownloadButton [Store Manager only]
├─ ContentArea
│   ├── PhotoGrid (default view)
│   │   └─ PhotoCard[] (repeating)
│   └─ PhotoList (alternate view)
│       └─ PhotoRow[] (repeating)
├─ Pagination
│   ├── ResultCount
│   └─ LoadMoreButton
└─ LightboxModal
    ├── PhotoViewer
    ├── PhotoInfoPanel
    └─ NavigationControls
```

3.2 Component Specifications

Component	Type	Description	Requirements
PageHeader	Container	Title and status summary	REQ-S003-UI-001
StatusSummary	Display	Shows counts by review status	REQ-S003-UI-002
FilterBar	Form	Filter controls for gallery	REQ-S003-UI-003

ViewToggle	Button Group	Grid/List view switch	REQ-S003-UI-004
PhotoGrid	Gallery	Thumbnail card grid	REQ-S003-UI-005
PhotoCard	Card	Individual photo thumbnail	REQ-S003-UI-006
PhotoList	Table	Tabular photo listing	REQ-S003-UI-007
LightboxModal	Modal	Full-size photo viewer	REQ-S003-UI-008
PhotoInfoPanel	Panel	Photo metadata display	REQ-S003-UI-009

3.3 Photo Card Layout

[Thumbnail]

✓

Window Poster

Summer Promo

Jun 15, 2025

by John D.

← Status overlay

← Item name

← Campaign name

← Upload date

← Uploader name

3.4 Status Overlay Specifications

Status	Icon	Background Color	Text Color
PENDING	⌚	amber-100	amber-800
APPROVED	✓	green-100	green-800
REJECTED	✗	red-100	red-800
SUPERSEDED	🔄	gray-100	gray-600

4. Data Requirements

4.1 API Endpoints

Endpoint	Method	Purpose	Request
/stores/{storeId}/photos	GET	Fetch photo gallery	Query params
/photos/{photoId}	GET	Get photo details	Path param
/photos/download	POST	Bulk download	Photo ID array

4.2 Request Parameters

GET /stores/{storeId}/photos

Parameter	Type	Required	Description
storeId	UUID	Yes	Store identifier (path)
campaign_id	UUID	No	Filter by campaign
status	Enum	No	PhotoReviewStatus filter
date_from	Date	No	Start date filter
date_to	Date	No	End date filter
item_type	Enum[]	No	Kit item type filter
uploaded_by	UUID	No	Uploader user ID
page	Integer	No	Page number (default: 1)
limit	Integer	No	Results per page (default: 24)

4.3 Response Schema

```
interface PhotoGalleryResponse {
  photos: PhotoUploadDTO[];
  meta: {
    total: number;
    page: number;
    limit: number;
    hasMore: boolean;
  };
  summary: {
    approved: number;
    pending: number;
    rejected: number;
    superseded: number;
  };
}

interface PhotoUploadDTO {
  id: string; // UUID
  file_url: string; // S3 signed URL
  thumbnail_url: string; // S3 signed thumbnail URL
  review_status: PhotoReviewStatus; // Enum
  created_at: string; // ISO 8601

  // Related data
  item_name: string; // KitItem.name
  item_type: ItemType; // KitItem.item_type
  slot_name: string | null; // LocationSlot.name
  campaign_id: string; // Campaign.id
  campaign_name: string; // Campaign.name
  uploaded_by: string; // User.id
  uploader_name: string; // User.name
}
```

```

// Review data (if reviewed)
rejection_reason?: RejectionReasonCode;
admin_comment?: string;
reviewed_at?: string;
reviewer_name?: string;

// Superseded link
superseded_by_id?: string;           // Replacement photo ID
}

```

4.4 Database Query

```

SELECT
  pu.id,
  pu.file_url,
  pu.thumbnail_url,
  pu.review_status,
  pu.created_at,
  ki.name as item_name,
  ki.item_type,
  ls.name as slot_name,
  c.id as campaign_id,
  c.name as campaign_name,
  u.id as uploaded_by,
  u.name as uploader_name,
  pr.rejection_reason,
  pr.admin_comment,
  pr.created_at as reviewed_at,
  rev.name as reviewer_name,
  pu.superseded_by_id
FROM photo_uploads pu
JOIN assignment_items ai ON pu.assignment_item_id = ai.id
JOIN kit_items ki ON ai.kit_item_id = ki.id
LEFT JOIN location_slots ls ON ai.location_slot_id = ls.id
JOIN store_assignments sa ON ai.store_assignment_id = sa.id
JOIN campaigns c ON sa.campaign_id = c.id
JOIN users u ON pu.uploaded_by = u.id
LEFT JOIN photo_reviews pr ON pr.photo_upload_id = pu.id
  AND pr.id = (SELECT id FROM photo_reviews WHERE photo_upload_id = pu.id ORDER BY
created_at DESC LIMIT 1)
LEFT JOIN users rev ON pr.reviewer_id = rev.id
WHERE sa.store_id = :storeId
  AND pu.deleted_at IS NULL
  AND sa.deleted_at IS NULL
ORDER BY pu.created_at DESC
LIMIT :limit OFFSET :offset

```

4.5 Caching Strategy

Data Type	Cache Duration	Invalidation Trigger
Photo list	5 minutes	New photo upload, status change
Photo thumbnails	24 hours	Photo superseded
Status summary	5 minutes	Any photo status change

5. Business Rules & Validation

5.1 Display Rules

Rule ID	Rule Description
REQ-S003-BR-001	Photos ordered by created_at descending (newest first)
REQ-S003-BR-002	Superseded photos displayed with gray overlay and link to replacement
REQ-S003-BR-003	Rejected photos show rejection reason and admin comment
REQ-S003-BR-004	Default view is Grid; user preference persisted in localStorage
REQ-S003-BR-005	Default filter shows photos from last 90 days

5.2 Filter Rules

Rule ID	Rule Description
REQ-S003-BR-006	Campaign filter shows only campaigns with photos for this store
REQ-S003-BR-007	Date filter options: Last 7 days, Last 30 days, Last 90 days, Custom range
REQ-S003-BR-008	Item type filter shows only types present in store's photos
REQ-S003-BR-009	Uploaded By filter visible only to Store Manager
REQ-S003-BR-010	Multiple filters combine with AND logic

5.3 Lightbox Rules

Rule ID	Rule Description
REQ-S003-BR-011	Lightbox opens on photo card click or keyboard Enter
REQ-S003-BR-012	Lightbox displays full-resolution image with zoom capability
REQ-S003-BR-013	Info panel shows all photo metadata and review details
REQ-S003-BR-014	Rejected photos show replacement photo preview if superseded
REQ-S003-BR-015	Navigation arrows cycle through filtered photo set

5.4 Download Rules

Rule ID	Rule Description
---------	------------------

REQ-S003-BR-016	Individual photo download available from lightbox
REQ-S003-BR-017	Bulk download requires Store Manager role
REQ-S003-BR-018	Bulk download creates ZIP archive with naming convention
REQ-S003-BR-019	Download filename format: {campaign}_{item}_{date}.{ext}

6. API Integration Points

6.1 Gallery Data Flow

```
sequenceDiagram
    participant U as Store User
    participant App as Store Portal
    participant API as Backend API
    participant S3 as AWS S3

    U->>App: Navigate to Photo Gallery
    App->>API: GET /stores/{storeId}/photos
    API->>API: Validate store membership
    API->>API: Query photo_uploads with joins
    API->>S3: Generate signed URLs (thumbnails)
    API-->>App: PhotoGalleryResponse
    App->>U: Render photo grid
```

6.2 Lightbox Data Flow

```
sequenceDiagram
    participant U as Store User
    participant App as Store Portal
    participant API as Backend API
    participant S3 as AWS S3

    U->>App: Click photo thumbnail
    App->>API: GET /photos/{photoId}
    API->>S3: Generate signed URL (full resolution)
    API-->>App: PhotoDetailDTO
    App->>U: Open lightbox with full image
    App->>S3: Load full-resolution image
```

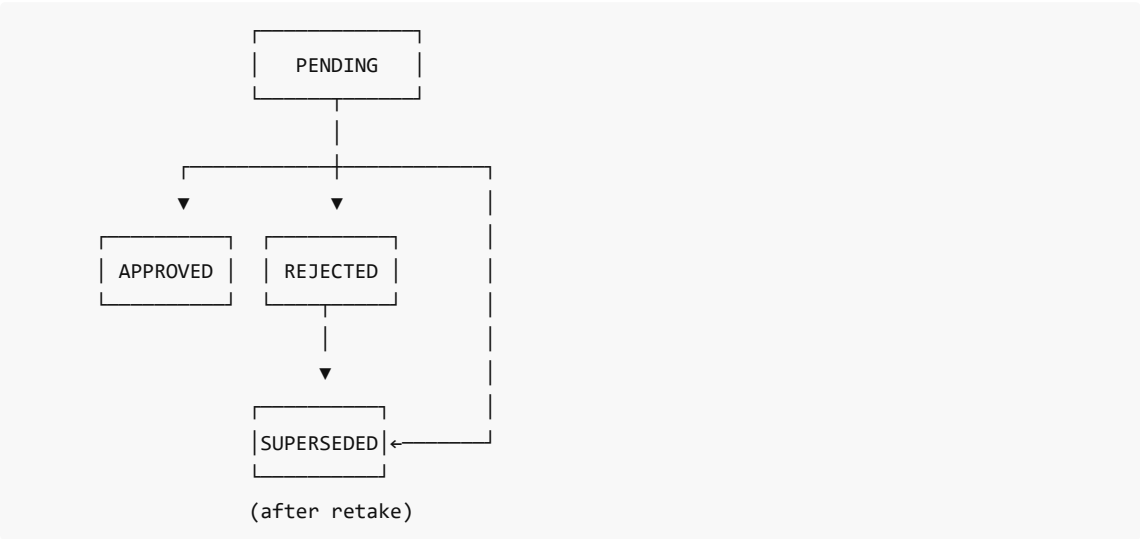
6.3 Integration Dependencies

System	Integration	Purpose
AWS S3	Signed URLs	Secure photo access
Photo Review Service	Status data	Review status and comments

Campaign Service	Campaign data	Campaign names and filters
------------------	---------------	----------------------------

7. State Transitions

7.1 Photo Review Status States



7.2 Status Transition Rules

From State	To State	Trigger	Actor
PENDING	APPROVED	Photo approved	Brand Admin, Campaign Manager, Regional Manager
PENDING	REJECTED	Photo rejected	Brand Admin, Campaign Manager, Regional Manager
REJECTED	SUPERSEDED	Retake photo uploaded	Store User
PENDING	SUPERSEDED	New photo replaces before review	Store User

7.3 View State Management

State	Persisted	Storage
View mode (Grid/List)	Yes	localStorage
Active filters	No	URL query params
Lightbox position	No	Component state
Scroll position	No	Component state

8. Error Handling

8.1 Error Scenarios

Error Code	Scenario	User Message	Recovery Action
ERR-S003-001	Photos API failure	"Unable to load photos. Please try again."	Retry button
ERR-S003-002	Photo not found	"Photo no longer available."	Remove from view, refresh
ERR-S003-003	Thumbnail load failure	Display placeholder	Auto-retry with fallback
ERR-S003-004	Signed URL expired	"Photo link expired."	Auto-refresh URL
ERR-S003-005	Download failure	"Download failed. Please try again."	Retry button
ERR-S003-006	Bulk download timeout	"Download is taking longer than expected."	Background job with notification
ERR-S003-007	Invalid filter combination	"No photos match your filters."	Clear filters option

8.2 Loading States

State	Display
Initial load	Skeleton grid with 24 placeholder cards
Filter change	Inline loading indicator
Load more	Loading spinner below grid
Lightbox loading	Blur backdrop with spinner
Download processing	Progress indicator with cancel option

8.3 Empty States

Condition	Message	Action
No photos for store	"No photos have been submitted yet."	Link to active campaigns
No photos match filters	"No photos match your current filters."	Clear filters button
All photos superseded	"All photos have been replaced."	Show superseded toggle

9. Accessibility Requirements

9.1 WCAG 2.1 AA Compliance

Requirement ID	Requirement	Implementation
REQ-S003-A11Y-001	Keyboard navigation	Full grid navigation with arrow keys
REQ-S003-A11Y-002	Focus management	Visible focus indicators on all interactive elements
REQ-S003-A11Y-003	Screen reader support	Alt text on all images, ARIA labels on controls
REQ-S003-A11Y-004	Lightbox keyboard control	Escape to close, arrows to navigate
REQ-S003-A11Y-005	Status announcements	Live region for filter result counts
REQ-S003-A11Y-006	Color independence	Status indicated by icon and text, not color alone

9.2 Keyboard Shortcuts

Key	Action	Context
Tab	Move focus between elements	Page
Enter	Open photo lightbox	Photo card focus
Escape	Close lightbox	Lightbox open
← / →	Navigate photos	Lightbox open
D	Download current photo	Lightbox open

9.3 Screen Reader Announcements

Trigger	Announcement
Gallery load	"Photo gallery loaded. {count} photos. {approved} approved, {pending} pending review."
Filter applied	"Showing {count} photos matching filters."
Lightbox open	"Photo viewer. {item_name} from {campaign_name}. Status: {status}. Use arrow keys to navigate."
Photo navigation	"Photo {current} of {total}. {item_name}. Status: {status}."

10. Acceptance Criteria

10.1 Functional Requirements

ID	Requirement	Priority	Verification
REQ-S003-FR-001	Gallery displays all photos for authenticated user's store	Must	Test
REQ-S003-FR-002	Grid view shows photo thumbnails with status overlay	Must	Test
REQ-S003-FR-003	List view displays tabular photo data	Must	Test

REQ-S003-FR-004	Campaign filter narrows results to selected campaign	Must	Test
REQ-S003-FR-005	Status filter shows only photos with selected status	Must	Test
REQ-S003-FR-006	Date filter limits results to date range	Must	Test
REQ-S003-FR-007	Clicking photo opens lightbox with full-size image	Must	Test
REQ-S003-FR-008	Lightbox displays complete photo metadata	Must	Test
REQ-S003-FR-009	Rejected photos show rejection reason and admin comment	Must	Test
REQ-S003-FR-010	Superseded photos link to replacement photo	Must	Test
REQ-S003-FR-011	Download button exports individual photo	Should	Test
REQ-S003-FR-012	Bulk download exports selected photos as ZIP	Should	Test
REQ-S003-FR-013	Lightbox navigation with arrow keys and buttons	Must	Test
REQ-S003-FR-014	Infinite scroll or load more pagination	Should	Test

10.2 Non-Functional Requirements

ID	Requirement	Target	Verification
REQ-S003-NFR-001	Gallery initial load < 2 seconds	P95	Performance test
REQ-S003-NFR-002	Thumbnail load < 500ms per image	P95	Performance test
REQ-S003-NFR-003	Full-size image load < 3 seconds	P95	Performance test
REQ-S003-NFR-004	Filter response < 1 second	P95	Performance test
REQ-S003-NFR-005	Support galleries with 500+ photos	Required	Load test
REQ-S003-NFR-006	Mobile responsive layout	Required	Visual test

10.3 Security Requirements

ID	Requirement	Verification
REQ-S003-SEC-001	Only authenticated store members can view gallery	Penetration test
REQ-S003-SEC-002	Cross-store photo access prevented	Security test
REQ-S003-SEC-003	S3 signed URLs expire after 1 hour	Configuration test
REQ-S003-SEC-004	Download requests logged in audit trail	Audit test

11. Traceability Matrix

Requirement	Source	SUPP Reference	Test Case
REQ-S003-FR-001	S03 Screen Spec	SUPP-018	TC-S003-001

REQ-S003-FR-002	S03 Screen Spec	SUPP-018	TC-S003-002
REQ-S003-FR-007	S03 Screen Spec	SUPP-018	TC-S003-007
REQ-S003-FR-009	S03 Screen Spec	SUPP-018	TC-S003-009
REQ-S003-SEC-001	Permission Matrix	SUPP-003	TC-S003-SEC-001

12. Related Screens

Screen	Relationship	Navigation
S001 Dashboard	Parent	Back link
S002 Campaign History	Sibling	Campaign filter
M05 Photo Capture	Input	Photo upload source
B07 Verification	Related	Brand review photos

Document Status: Complete IEEE 830 Compliance: Section 3.2 - Specific Requirements / Functional Requirements

S004 Team Management - Screen Specification

SRS Section: 5.9.4 | **Module:** Store Portal | **Version:** 1.0 **IEEE 830 Reference:** Section 3.2 - Functional Requirements
Source Documents:

- [S04 Team Management Screen Spec](#)
- [SUPP-001 Personas](#)
- [SUPP-003 RBAC](#) **Last Updated:** 2026-01-01

1. Screen Overview

1.1 Purpose

The Team Management screen enables Store Managers to administer their store's team members, including inviting new users, managing roles, tracking activity metrics, and removing access. This screen is restricted to Store Manager role only.

1.2 Route Configuration

Attribute	Value
Route Path	/store/team
Route Type	Protected (Authentication Required)
Role Restriction	STORE_MANAGER only
Lazy Loading	Yes

Mobile Support	Full Responsive
----------------	-----------------

1.3 Screen Context

Attribute	Description
Primary Purpose	Manage store team membership and permissions
Entry Points	Store Dashboard navigation, Dashboard team widget
Exit Points	Dashboard, Reports
Session Scope	Store context from authenticated membership

1.4 Screenshot Reference

 Team Management Screen

2. User Roles & Permissions

2.1 Authorized Roles

Role	Access Level	Restrictions
STORE_MANAGER (P07)	Full Access	Own store team only
STORE_OPERATOR (P08)	No Access	Screen not visible

2.2 Permission Requirements

Permission	STORE_MANAGER	STORE_OPERATOR
View team list	Y	N
Invite team members	Y	N
Edit member role	Y	N
Deactivate members	Y	N
Remove members	Y	N
View activity metrics	Y	N
Resend invitations	Y	N
Cancel invitations	Y	N

2.3 Data Scoping Rules

Rule ID	Description
REQ-S004-SEC-001	Team members filtered to authenticated user's store via memberships.store_id

REQ-S004-SEC-002	Store Manager role validated before screen access
REQ-S004-SEC-003	Cross-store membership modification prevented at API layer
REQ-S004-SEC-004	Self-removal prevented to maintain store manager continuity

3. UI Components

3.1 Component Hierarchy



3.2 Component Specifications

Component	Type	Description	Requirements
PageHeader	Container	Title and invite button	REQ-S004-UI-001
MemberTable	Data Table	Active team members	REQ-S004-UI-002
InvitationTable	Data Table	Pending invitations	REQ-S004-UI-003
ActivityTable	Data Table	30-day activity metrics	REQ-S004-UI-004
InviteMemberModal	Modal Dialog	New member invitation form	REQ-S004-UI-005
EditMemberModal	Modal Dialog	Member role/status editor	REQ-S004-UI-006
RoleBadge	Badge	Manager/User role indicator	REQ-S004-UI-007
StatusBadge	Badge	Active/Invited/Inactive status	REQ-S004-UI-008

3.3 Role Badge Specifications

Role	Label	Color
STORE_MANAGER	"Manager"	blue-100 / blue-800
STORE_OPERATOR	"User"	gray-100 / gray-700

3.4 Status Badge Specifications

Status	Label	Color	Description
Active	"Active"	green-100 / green-800	Registered and active
Invited	"Invited"	amber-100 / amber-800	Pending registration
Inactive	"Inactive"	gray-100 / gray-600	Deactivated by manager

4. Data Requirements

4.1 API Endpoints

Endpoint	Method	Purpose	Authorization
/stores/{storeId}/members	GET	List team members	STORE_MANAGER
/stores/{storeId}/invitations	GET	List pending invitations	STORE_MANAGER
/stores/{storeId}/invitations	POST	Create invitation	STORE_MANAGER
/stores/{storeId}/invitations/{id}/resend	POST	Resend invitation email	STORE_MANAGER
/stores/{storeId}/invitations/{id}	DELETE	Cancel invitation	STORE_MANAGER
/memberships/{id}	PATCH	Update member role/status	STORE_MANAGER

/memberships/{id}	DELETE	Remove member from store	STORE_MANAGER
/stores/{storeId}/activity	GET	Team activity summary	STORE_MANAGER

4.2 Request/Response Schemas

GET /stores/{storeId}/members Response

```
interface TeamMembersResponse {
  members: TeamMemberDTO[];
  meta: {
    total: number;
    managerCount: number;
    operatorCount: number;
  };
}

interface TeamMemberDTO {
  id: string;           // Membership ID
  user_id: string;      // User ID
  name: string;         // User full name
  email: string;        // User email
  phone?: string;       // User phone
  avatar_url?: string;  // Profile picture URL
  role: 'STORE_MANAGER' | 'STORE_OPERATOR';
  status: 'active' | 'inactive';
  joined_at: string;    // ISO 8601
  last_active_at?: string; // ISO 8601
  photo_count: number;  // Total photos uploaded
  is_current_user: boolean; // Flag for self
}
```

GET /stores/{storeId}/invitations Response

```
interface InvitationsResponse {
  invitations: InvitationDTO[];
}

interface InvitationDTO {
  id: string;           // Invitation ID
  email: string;        // Invited email
  role: 'STORE_MANAGER' | 'STORE_OPERATOR';
  invited_at: string;    // ISO 8601
  expires_at: string;    // ISO 8601
  invited_by: string;    // Inviter user name
  status: 'pending' | 'expired';
}
```

POST /stores/{storeId}/invitations Request

```
interface CreateInvitationRequest {
    email: string;           // Required
    role: 'STORE_MANAGER' | 'STORE_OPERATOR'; // Required
    message?: string;       // Optional personal message
}
```

GET /stores/{storeId}/activity Response

```
interface TeamActivityResponse {
    activity: TeamMemberActivityDTO[];
    period: {
        from: string;           // ISO 8601
        to: string;             // ISO 8601
        days: number;           // Default 30
    };
}

interface TeamMemberActivityDTO {
    user_id: string;
    user_name: string;
    photos: number;             // Photos uploaded
    receipts: number;           // Receipt surveys completed
    installs: number;           // Install surveys completed
    issues: number;             // Issues reported
}
```

4.3 Database Query - Team Members

```
SELECT
    m.id as membership_id,
    u.id as user_id,
    u.name,
    u.email,
    u.phone,
    u.avatar_url,
    m.role,
    CASE WHEN u.is_active THEN 'active' ELSE 'inactive' END as status,
    m.created_at as joined_at,
    u.last_active_at,
    COUNT(pu.id) as photo_count
FROM memberships m
JOIN users u ON m.user_id = u.id
LEFT JOIN photo_uploads pu ON pu.uploaded_by = u.id
WHERE m.store_id = :storeId
    AND m.deleted_at IS NULL
    AND u.deleted_at IS NULL
GROUP BY m.id, u.id
ORDER BY
```

```
CASE m.role WHEN 'STORE_MANAGER' THEN 0 ELSE 1 END,
u.name
```

4.4 Database Query - Team Activity

```
WITH activity_period AS (
  SELECT
    CURRENT_DATE - INTERVAL '30 days' as start_date,
    CURRENT_DATE as end_date
)
SELECT
  u.id as user_id,
  u.name as user_name,
  COUNT(DISTINCT pu.id) as photos,
  COUNT(DISTINCT CASE WHEN ssr.survey_type = 'receipt' THEN ssr.id END) as receipts,
  COUNT(DISTINCT CASE WHEN ssr.survey_type = 'install' THEN ssr.id END) as installs,
  COUNT(DISTINCT ir.id) as issues
FROM memberships m
JOIN users u ON m.user_id = u.id
CROSS JOIN activity_period ap
LEFT JOIN photo_uploads pu ON pu.uploaded_by = u.id
  AND pu.created_at BETWEEN ap.start_date AND ap.end_date
LEFT JOIN store_survey_responses ssr ON ssr.submitted_by = u.id
  AND ssr.submitted_at BETWEEN ap.start_date AND ap.end_date
LEFT JOIN issue_requests ir ON ir.reported_by = u.id
  AND ir.created_at BETWEEN ap.start_date AND ap.end_date
WHERE m.store_id = :storeId
  AND m.deleted_at IS NULL
GROUP BY u.id, u.name
ORDER BY u.name
```

5. Business Rules & Validation

5.1 Invitation Rules

Rule ID	Rule Description
REQ-S004-BR-001	Email must be valid format and not already a member of this store
REQ-S004-BR-002	Email must not have pending invitation for this store
REQ-S004-BR-003	Invitations expire after 7 days
REQ-S004-BR-004	Maximum 5 pending invitations per store at a time
REQ-S004-BR-005	Resend resets expiration date to 7 days from now
REQ-S004-BR-006	Personal message limited to 500 characters

5.2 Role Management Rules

Rule ID	Rule Description
REQ-S004-BR-007	Store must always have at least one active STORE_MANAGER
REQ-S004-BR-008	Cannot demote last active manager to STORE_OPERATOR
REQ-S004-BR-009	Cannot deactivate or remove last active manager
REQ-S004-BR-010	Cannot remove self (current user) from store
REQ-S004-BR-011	Role changes take effect immediately

5.3 Status Management Rules

Rule ID	Rule Description
REQ-S004-BR-012	Deactivating user revokes API access immediately
REQ-S004-BR-013	Deactivated users retain data but cannot log in
REQ-S004-BR-014	Reactivation restores all previous permissions
REQ-S004-BR-015	Removed users are soft-deleted from membership

5.4 Activity Calculation Rules

Rule ID	Rule Description
REQ-S004-BR-016	Activity period is rolling 30 days from current date
REQ-S004-BR-017	Photo count includes all statuses (pending, approved, rejected)
REQ-S004-BR-018	Survey counts based on submitted_at timestamp
REQ-S004-BR-019	Issue count includes all issue statuses

6. API Integration Points

6.1 Load Team Data Flow

```
sequenceDiagram
    participant U as Store Manager
    participant App as Store Portal
    participant API as Backend API

    U->>App: Navigate to Team Management
    App->>API: GET /stores/{storeId}/members
    App->>API: GET /stores/{storeId}/invitations
    App->>API: GET /stores/{storeId}/activity
    API-->>App: TeamMembersResponse
    API-->>App: InvitationsResponse
```

```
API-->>App: TeamActivityResponse
App->>U: Render team tables
```

6.2 Invite Member Flow

```
sequenceDiagram
    participant U as Store Manager
    participant App as Store Portal
    participant API as Backend API
    participant Email as Email Service

    U->>App: Click "Invite Member"
    App->>U: Show invite modal
    U->>App: Enter email, select role, add message
    U->>App: Click "Send Invitation"
    App->>API: POST /stores/{storeId}/invitations
    API->>API: Validate email uniqueness
    API->>API: Create store_invitation record
    API->>Email: Send invitation email
    API-->>App: InvitationDTO
    App->>U: Show success toast, update invitations table
```

6.3 Edit Member Flow

```
sequenceDiagram
    participant U as Store Manager
    participant App as Store Portal
    participant API as Backend API

    U->>App: Click member row
    App->>U: Show edit modal
    U->>App: Change role or status
    U->>App: Click "Save"
    App->>API: PATCH /memberships/{id}
    API->>API: Validate last manager rule
    API->>API: Update membership
    API-->>App: Updated TeamMemberDTO
    App->>U: Update table, show success toast
```

6.4 Remove Member Flow

```
sequenceDiagram
    participant U as Store Manager
    participant App as Store Portal
    participant API as Backend API

    U->>App: Click "Remove from Store"
    App->>U: Show confirmation dialog
```

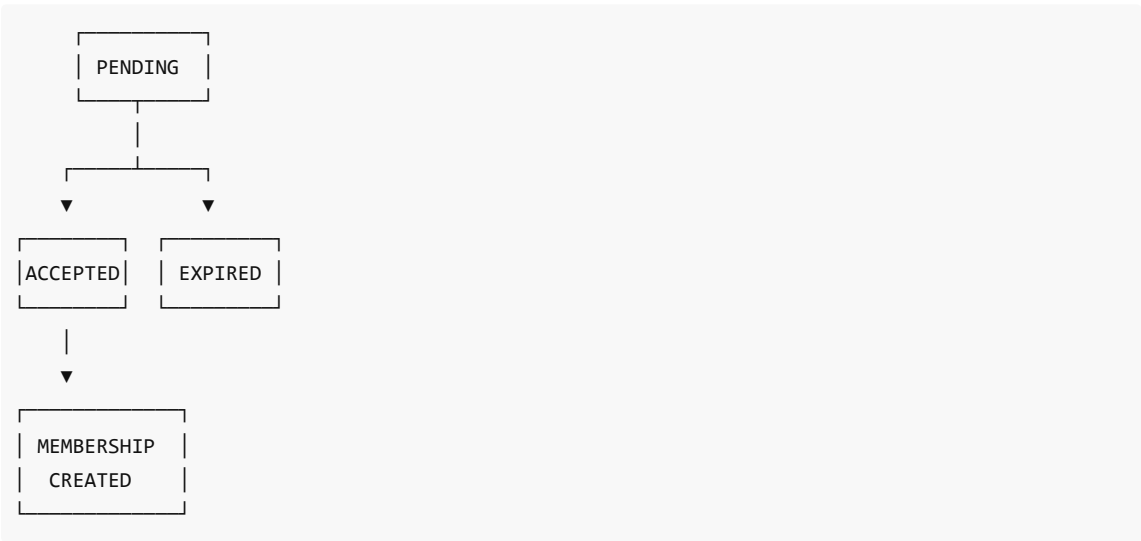
```
U->>App: Confirm removal
App->>API: DELETE /memberships/{id}
API->>API: Validate not last manager
API->>API: Soft delete membership
API-->>App: 204 No Content
App->>U: Remove from table, show success toast
```

6.5 Integration Dependencies

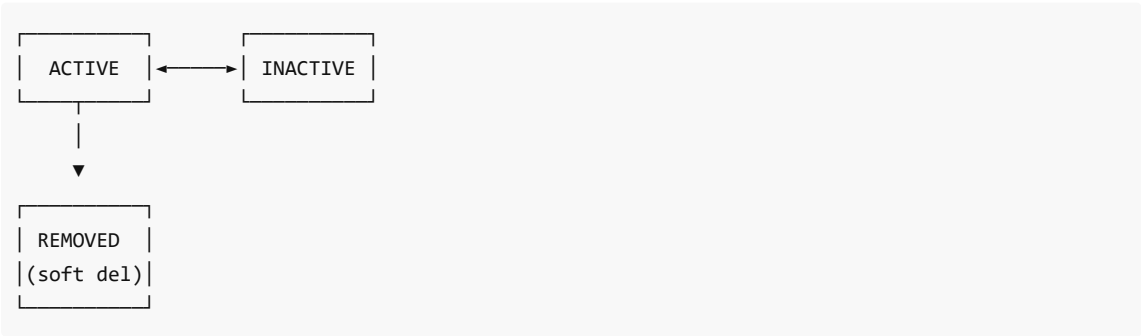
System	Integration	Purpose
Email Service	SMTP/SendGrid	Invitation delivery
Auth Service	JWT Validation	Permission verification
Audit Service	Event logging	Track team changes

7. State Transitions

7.1 Invitation Status States



7.2 Member Status States



7.3 State Transition Actions

From	To	Action	Actor
-	PENDING	Create invitation	Store Manager
PENDING	ACCEPTED	User accepts invite	Invited User
PENDING	EXPIRED	Time passes 7 days	System
PENDING	CANCELLED	Cancel invitation	Store Manager
ACTIVE	INACTIVE	Deactivate member	Store Manager
INACTIVE	ACTIVE	Reactivate member	Store Manager
ACTIVE/INACTIVE	REMOVED	Remove member	Store Manager

8. Error Handling

8.1 Error Scenarios

Error Code	Scenario	User Message	Recovery Action
ERR-S004-001	Team API failure	"Unable to load team members. Please try again."	Retry button
ERR-S004-002	Email already member	"This email is already a member of your store."	Clear email field
ERR-S004-003	Email has pending invite	"An invitation has already been sent to this email."	Offer resend
ERR-S004-004	Invalid email format	"Please enter a valid email address."	Inline validation
ERR-S004-005	Last manager removal	"Cannot remove the last manager. Promote another member first."	Dismiss dialog
ERR-S004-006	Self-removal attempt	"You cannot remove yourself from the store."	Dismiss dialog
ERR-S004-007	Invitation send failed	"Unable to send invitation. Please try again."	Retry button
ERR-S004-008	Invitation limit reached	"Maximum pending invitations reached. Cancel existing invites first."	Show pending list
ERR-S004-009	Role update failed	"Unable to update role. Please try again."	Retry button
ERR-S004-010	Member not found	"Team member no longer exists."	Refresh table

8.2 Validation Messages

Field	Validation	Message
Email	Required	"Email address is required"
Email	Format	"Please enter a valid email address"
Email	Duplicate	"This email is already on your team"
Role	Required	"Please select a role"
Message	Max length	"Message cannot exceed 500 characters"

8.3 Loading States

State	Display
Initial load	Skeleton table with 5 rows
Invite submitting	"Sending invitation..." with spinner
Role updating	Inline spinner on affected row
Member removing	Confirmation dialog with spinner

8.4 Empty States

Condition	Message	Action
No team members (impossible)	N/A	Current user always shown
No pending invitations	"No pending invitations"	Invite button
No activity in 30 days	"No activity recorded in the last 30 days"	None

9. Accessibility Requirements

9.1 WCAG 2.1 AA Compliance

Requirement ID	Requirement	Implementation
REQ-S004-A11Y-001	Keyboard navigation	Full table navigation with Tab/Arrow keys
REQ-S004-A11Y-002	Focus management	Modal focus trap, return focus on close
REQ-S004-A11Y-003	Screen reader support	ARIA labels, role announcements
REQ-S004-A11Y-004	Form accessibility	Associated labels, error announcements
REQ-S004-A11Y-005	Confirmation dialogs	Focus on primary action, Escape to close
REQ-S004-A11Y-006	Status announcements	Live region for success/error toasts

9.2 Keyboard Shortcuts

Key	Action	Context
-----	--------	---------

Tab	Move focus between elements	Page
Enter	Open edit modal	Table row focus
Escape	Close modal	Modal open
Delete	Open remove confirmation	Table row focus

9.3 Screen Reader Announcements

Trigger	Announcement
Page load	"Team Management. {count} active members, {count} pending invitations."
Invite success	"Invitation sent to {email}."
Role updated	"{name} role changed to {role}."
Member removed	"{name} has been removed from the team."
Error	Error message content

10. Acceptance Criteria

10.1 Functional Requirements

ID	Requirement	Priority	Verification
REQ-S004-FR-001	Screen restricted to STORE_MANAGER role	Must	Test
REQ-S004-FR-002	Team list displays all active members with roles	Must	Test
REQ-S004-FR-003	Pending invitations displayed in separate section	Must	Test
REQ-S004-FR-004	Invite modal captures email and role selection	Must	Test
REQ-S004-FR-005	Invitation email sent upon invite creation	Must	Test
REQ-S004-FR-006	Manager can change member role	Must	Test
REQ-S004-FR-007	Manager can deactivate member	Must	Test
REQ-S004-FR-008	Manager can remove member from store	Must	Test
REQ-S004-FR-009	Cannot remove last active manager	Must	Test
REQ-S004-FR-010	Cannot remove self from store	Must	Test
REQ-S004-FR-011	Activity summary shows 30-day metrics	Should	Test
REQ-S004-FR-012	Resend option for pending invitations	Should	Test
REQ-S004-FR-013	Cancel option for pending invitations	Should	Test
REQ-S004-FR-014	Last active date shown for each member	Should	Test

10.2 Non-Functional Requirements

ID	Requirement	Target	Verification
REQ-S004-NFR-001	Page load < 2 seconds	P95	Performance test
REQ-S004-NFR-002	Invite creation < 3 seconds	P95	Performance test
REQ-S004-NFR-003	Role update < 1 second	P95	Performance test
REQ-S004-NFR-004	Support teams up to 50 members	Required	Load test
REQ-S004-NFR-005	Email delivery within 5 minutes	P95	Integration test

10.3 Security Requirements

ID	Requirement	Verification
REQ-S004-SEC-001	Only STORE_MANAGER can access screen	Authorization test
REQ-S004-SEC-002	Cross-store member manipulation prevented	Security test
REQ-S004-SEC-003	All team changes logged to audit trail	Audit test
REQ-S004-SEC-004	Email invitation tokens expire after 7 days	Configuration test
REQ-S004-SEC-005	Invitation tokens are single-use	Security test

11. Traceability Matrix

Requirement	Source	SUPP Reference	Test Case
REQ-S004-FR-001	S04 Screen Spec	SUPP-003	TC-S004-001
REQ-S004-FR-002	S04 Screen Spec	SUPP-001	TC-S004-002
REQ-S004-FR-005	S04 Screen Spec	SUPP-003	TC-S004-005
REQ-S004-FR-009	S04 Screen Spec	SUPP-003	TC-S004-009
REQ-S004-SEC-001	Permission Matrix	SUPP-003	TC-S004-SEC-001

12. Related Screens

Screen	Relationship	Navigation
S001 Dashboard	Parent	Team widget shows summary
S005 Reports	Sibling	Detailed team analytics
M07 Profile	Related	User self-service

S005 Store Reports - Screen Specification

SRS Section: 5.9.5 | **Version:** 1.0 | **Status:** Draft **Module:** Store Portal **Route:** /store/reports **Source:** [S05_Reports.md](#) **Last Updated:** 2026-01-01

1. Screen Overview

1.1 Purpose

The Store Reports screen provides Store Managers with comprehensive analytics and performance insights for their store's campaign execution, photo compliance, team performance, and issue resolution. This read-only analytics interface aggregates data across multiple entities to deliver actionable metrics with trend analysis and benchmarking capabilities.

1.2 Scope

This specification defines the functional requirements, data requirements, and user interface components for the Store Reports screen within the Store Portal module of NewPOPSys v1.

1.3 Primary Functions

Function	Description
View KPI Summary	Display key performance indicators with trend indicators
Analyze Campaigns	Review campaign completion rates and timing metrics
Review Photo Metrics	Examine approval rates and rejection patterns
Monitor Team Performance	Track individual team member contributions
Track Issues	Analyze issue patterns and resolution times
Export Reports	Download report data in multiple formats

1.4 Screen Context

Store Portal Navigation:

└─ Dashboard (S001)

└─ Campaign History (S002)

└─ Photo Gallery (S003)

└─ Team Management (S004)

└─ Reports (S005) ← Current Screen

- └─ Overview Tab
- └─ Campaigns Tab
- └─ Photos Tab
- └─ Team Tab
- └─ Issues Tab

2. User Roles & Permissions

2.1 Authorized Roles

Role	Access Level	Restrictions
STORE_MANAGER (P07)	Full Access	Own store data only
STORE_OPERATOR (P08)	No Access	Cannot access reports screen
REGIONAL_MANAGER (P06)	View Only	Via impersonation or region-scoped access
BRAND_ADMIN (P04)	View Only	Via impersonation
PLATFORM_ADMIN (P01)	Full Access	Via impersonation

2.2 Permission Requirements

Requirement ID	Permission	Description
REQ-S005-SEC-001	reports:read	View store reports and analytics
REQ-S005-SEC-002	reports:export	Download report data in various formats
REQ-S005-SEC-003	store:read	Access store-level aggregated metrics

2.3 Access Control Rules

REQ-S005-SEC-004: Data Isolation
- Users SHALL only view reports for stores where membership.store_id matches
- All API calls SHALL include tenant_id and store_id scoping
- Report data SHALL NOT include cross-store comparisons without aggregation
REQ-S005-SEC-005: Role Enforcement
- Store Operators SHALL be denied access with HTTP 403
- Navigation item SHALL be hidden for unauthorized roles
- Direct URL access SHALL redirect to Dashboard with access denied message

3. UI Components

3.1 Component Hierarchy

ReportsScreen
├ PageHeader
├ Title ("Store Reports")
├ DateRangePicker
└ ExportButton
├ TabNavigation
├ OverviewTab
├ CampaignsTab
├ PhotosTab
├ TeamTab
└ IssuesTab
└ TabContent

- | └─ KPICardGrid (4 cards)
- | └─ TrendChart (line chart)
- | └─ BreakdownCharts (pie, bar)
- | └─ DataTable (detailed data)
- └─ LoadingState / EmptyState

3.2 Component Specifications

3.2.1 Page Header

Element	Specification
Title	"Store Reports" - H1, left-aligned
Date Range Picker	Dropdown with preset options + custom range
Export Button	Primary action, right-aligned, opens format menu

3.2.2 Tab Navigation

Requirement ID	Requirement
REQ-S005-UI-001	Tab bar SHALL display 5 tabs: Overview, Campaigns, Photos, Team, Issues
REQ-S005-UI-002	Active tab SHALL be visually distinguished with underline indicator
REQ-S005-UI-003	Tab switching SHALL preserve selected date range
REQ-S005-UI-004	URL SHALL update to reflect active tab (e.g., /store/reports?tab=photos)

3.2.3 KPI Cards

Card	Metric	Threshold Colors
Compliance Rate	completed / total campaigns x 100	Green >= 90%, Yellow >= 75%, Red < 75%
On-Time Rate	on_time / completed x 100	Green >= 85%, Yellow >= 70%, Red < 70%
Photo Approval	approved / total photos x 100	Green >= 95%, Yellow >= 85%, Red < 85%
Avg Completion Time	avg(completed_at - start_date)	Lower is better (days)

Requirement ID	Requirement
REQ-S005-UI-005	KPI cards SHALL display current value and trend indicator (arrow up/down)
REQ-S005-UI-006	Trend SHALL compare to previous period of same duration
REQ-S005-UI-007	Cards SHALL use color coding based on threshold values
REQ-S005-UI-008	Hover state SHALL show exact values and period comparison

3.2.4 Charts

Chart Type	Tab	Data Displayed
------------	-----	----------------

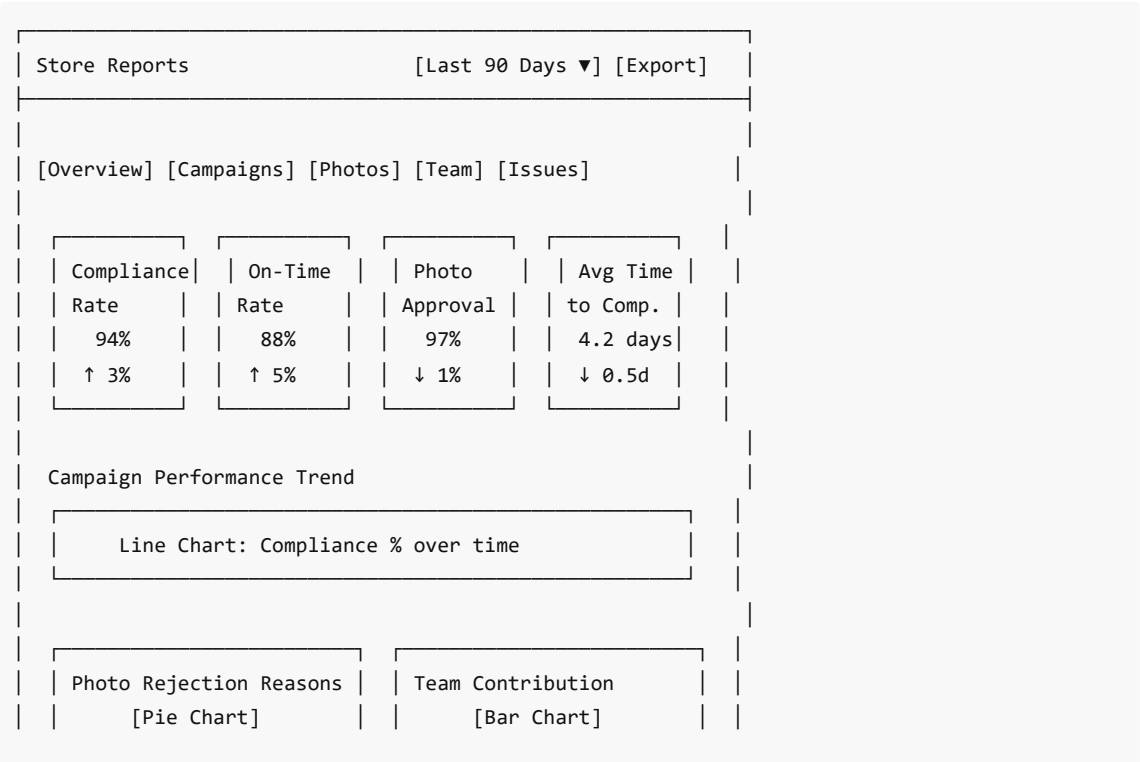
Line Chart	Overview	Campaign performance trend over time
Pie Chart	Overview, Photos	Rejection reasons breakdown
Bar Chart	Overview, Team	Team contribution percentages
Stacked Bar	Campaigns	On-time vs late completion by month

Requirement ID	Requirement
REQ-S005-UI-009	Charts SHALL be responsive and scale to container width
REQ-S005-UI-010	Charts SHALL display tooltips with exact values on hover
REQ-S005-UI-011	Charts SHALL include legends when multiple series displayed
REQ-S005-UI-012	Empty charts SHALL display "No data for selected period" message

3.2.5 Data Tables

Requirement ID	Requirement
REQ-S005-UI-013	Tables SHALL be sortable by clicking column headers
REQ-S005-UI-014	Tables SHALL display pagination for > 10 rows
REQ-S005-UI-015	Tables SHALL support column-specific formatting (% , dates, status badges)
REQ-S005-UI-016	Row click SHALL navigate to related detail screen where applicable

3.3 Reports Layout



Recent Campaigns				
Campaign	Completed	On-Time	Photos	Issues

4. Data Requirements

4.1 Data Model References

Entity	Source Table	Purpose
Store Assignment	store_assignments	Campaign completion metrics
Campaign	campaigns	Campaign details, dates
Photo Upload	photo_uploads	Photo count, approval rates
Photo Review	photo_reviews	Rejection reasons
Issue Request	issue_requests	Issue counts, resolution
User	users	Team member identification
Membership	memberships	Team member association

4.2 Database Queries

4.2.1 Compliance Metrics Query

```
SELECT
    COUNT(*) as total_campaigns,
    COUNT(CASE WHEN sa.status = 'COMPLETE' THEN 1 END) as completed,
    COUNT(CASE WHEN sa.status = 'COMPLETE'
        AND sa.completed_at <= c.install_end_date THEN 1 END) as on_time,
    AVG(EXTRACT(EPOCH FROM (sa.completed_at - c.install_start_date))) / 86400
        as avg_duration_days
FROM store_assignments sa
JOIN campaigns c ON sa.campaign_id = c.id
WHERE sa.store_id = :storeId
    AND sa.deleted_at IS NULL
    AND c.deleted_at IS NULL
    AND sa.created_at >= :startDate
    AND sa.created_at <= :endDate;
```

4.2.2 Photo Metrics Query

```
SELECT
    COUNT(pu.id) as total_photos,
    COUNT(CASE WHEN pu.review_status = 'APPROVED' THEN 1 END) as approved,
```

```

COUNT(CASE WHEN pu.review_status = 'REJECTED' THEN 1 END) as rejected,
COUNT(CASE WHEN pu.review_status = 'PENDING' THEN 1 END) as pending
FROM photo_uploads pu
JOIN assignment_items ai ON pu.assignment_item_id = ai.id
JOIN store_assignments sa ON ai.store_assignment_id = sa.id
WHERE sa.store_id = :storeId
AND pu.deleted_at IS NULL
AND pu.created_at >= :startDate
AND pu.created_at <= :endDate;

```

4.2.3 Rejection Reasons Query

```

SELECT
    pr.reason_code,
    COUNT(*) as count
FROM photo_reviews pr
JOIN photo_uploads pu ON pr.photo_upload_id = pu.id
JOIN assignment_items ai ON pu.assignment_item_id = ai.id
JOIN store_assignments sa ON ai.store_assignment_id = sa.id
WHERE sa.store_id = :storeId
AND pr.decision = 'REJECTED'
AND pr.deleted_at IS NULL
AND pr.created_at >= :startDate
AND pr.created_at <= :endDate
GROUP BY pr.reason_code
ORDER BY count DESC;

```

4.2.4 Team Contribution Query

```

SELECT
    u.id as user_id,
    u.first_name || ' ' || LEFT(u.last_name, 1) || '.' as display_name,
    COUNT(DISTINCT pu.id) as photos_uploaded,
    COUNT(DISTINCT CASE WHEN sa.status = 'COMPLETE' THEN sa.id END) as completions
FROM users u
JOIN memberships m ON u.id = m.user_id
LEFT JOIN photo_uploads pu ON pu.uploaded_by = u.id
LEFT JOIN assignment_items ai ON pu.assignment_item_id = ai.id
LEFT JOIN store_assignments sa ON ai.store_assignment_id = sa.id
WHERE m.store_id = :storeId
AND m.deleted_at IS NULL
AND u.is_active = true
AND (pu.created_at >= :startDate OR pu.created_at IS NULL)
AND (pu.created_at <= :endDate OR pu.created_at IS NULL)
GROUP BY u.id, u.first_name, u.last_name
ORDER BY photos_uploaded DESC;

```

4.2.5 Issues Summary Query

```

SELECT
    ir.issue_type,
    ir.status,
    COUNT(*) as count,
    AVG(EXTRACT(EPOCH FROM (ir.resolved_at - ir.created_at)) / 3600)
        as avg_resolution_hours
FROM issue_requests ir
JOIN store_assignments sa ON ir.store_assignment_id = sa.id
WHERE sa.store_id = :storeId
    AND ir.deleted_at IS NULL
    AND ir.created_at >= :startDate
    AND ir.created_at <= :endDate
GROUP BY ir.issue_type, ir.status
ORDER BY count DESC;

```

4.3 API Response Interfaces

```

interface StoreReportsResponse {
    storeId: string;
    dateRange: {
        start: string; // ISO date
        end: string; // ISO date
        label: string; // "Last 90 Days"
    };
    kpis: KPIMetrics;
    campaignMetrics: CampaignMetrics;
    photoMetrics: PhotoMetrics;
    teamMetrics: TeamMemberMetrics[];
    issueMetrics: IssueMetrics;
    trends: TrendData[];
}

interface KPIMetrics {
    complianceRate: {
        value: number; // Percentage (0-100)
        trend: number; // Change from previous period
        status: 'green' | 'yellow' | 'red';
    };
    onTimeRate: {
        value: number;
        trend: number;
        status: 'green' | 'yellow' | 'red';
    };
    photoApprovalRate: {
        value: number;
        trend: number;
        status: 'green' | 'yellow' | 'red';
    };
    avgCompletionDays: {
        value: number; // Days
    };
}

```

```
        trend: number;
        status: 'green' | 'yellow' | 'red';
    };
}
```

```
interface CampaignMetrics {
    total: number;
    completed: number;
    onTime: number;
    late: number;
    inProgress: number;
    campaigns: CampaignDetail[];
}
```

```
interface CampaignDetail {
    campaignId: string;
    campaignName: string;
    status: string;
    completedAt: string | null;
    onTime: boolean | null;
    photoCount: number;
    issueCount: number;
}
```

```
interface PhotoMetrics {
    total: number;
    approved: number;
    rejected: number;
    pending: number;
    approvalRate: number;
    rejectionReasons: RejectionReason[];
}
```

```
interface RejectionReason {
    code: string;
    label: string;
    count: number;
    percentage: number;
}
```

```
interface TeamMemberMetrics {
    userId: string;
    displayName: string;
    photosUploaded: number;
    completions: number;
    contributionPercentage: number;
}
```

```
interface IssueMetrics {
    total: number;
    open: number;
    resolved: number;
}
```

```
    avgResolutionHours: number;
    byType: IssueTypeBreakdown[];
}

interface IssueTypeBreakdown {
    type: string;
    count: number;
    percentage: number;
}

interface TrendData {
    period: string;           // "2025-12", "2025-W52"
    complianceRate: number;
    onTimeRate: number;
    photoApprovalRate: number;
}
```

4.4 Data Freshness

Requirement ID	Requirement
REQ-S005-DATA-001	Report data SHALL be aggregated in real-time (not cached)
REQ-S005-DATA-002	Date range filter SHALL apply to all metrics on screen
REQ-S005-DATA-003	Trend calculations SHALL compare equal-length periods
REQ-S005-DATA-004	Empty periods SHALL display zero values, not null

5. Business Rules & Validation

5.1 Date Range Rules

Requirement ID	Rule
REQ-S005-BR-001	Default date range SHALL be "Last 90 Days"
REQ-S005-BR-002	Maximum date range SHALL be 365 days
REQ-S005-BR-003	End date SHALL NOT exceed current date
REQ-S005-BR-004	Start date SHALL NOT be earlier than store creation date
REQ-S005-BR-005	Custom date range SHALL validate start < end

5.2 KPI Calculation Rules

Requirement ID	Rule
REQ-S005-BR-006	Compliance Rate = (completed campaigns / total assigned campaigns) x 100
REQ-S005-BR-007	On-Time Rate = (on-time completions / completed campaigns) x 100

REQ-S005-BR-008	Photo Approval Rate = (approved photos / total reviewed photos) x 100
REQ-S005-BR-009	Avg Completion Time = average of (completed_at - install_start_date) in days
REQ-S005-BR-010	Trend SHALL be calculated as (current_period - previous_period)

5.3 Threshold Rules

KPI	Green	Yellow	Red
Compliance Rate	>= 90%	>= 75% and < 90%	< 75%
On-Time Rate	>= 85%	>= 70% and < 85%	< 70%
Photo Approval	>= 95%	>= 85% and < 95%	< 85%
Avg Completion	Contextual	Contextual	Contextual

Requirement ID	Rule
REQ-S005-BR-011	KPI status colors SHALL be applied based on threshold table
REQ-S005-BR-012	Avg Completion Time status SHALL compare to network average

5.4 Data Scope Rules

Requirement ID	Rule
REQ-S005-BR-013	Reports SHALL only include store_assignments with status != 'CANCELLED'
REQ-S005-BR-014	Photo counts SHALL exclude deleted photos (deleted_at IS NOT NULL)
REQ-S005-BR-015	Team metrics SHALL only include active users (is_active = true)
REQ-S005-BR-016	Issue counts SHALL include all statuses except 'DENIED'

5.5 Comparison Rules

Requirement ID	Rule
REQ-S005-BR-017	Network average SHALL be calculated across all stores in same brand
REQ-S005-BR-018	Region average SHALL be calculated across stores in same region
REQ-S005-BR-019	Period-over-period SHALL compare equivalent duration periods

6. API Integration Points

6.1 Endpoint Specifications

6.1.1 Get Store Reports

```
GET /api/v1/stores/{storeId}/reports
```

Query Parameters:

Parameter	Type	Required	Description
range	string	No	Preset range: 7d, 30d, 90d, 365d (default: 90d)
startDate	string	No	Custom start date (ISO 8601)
endDate	string	No	Custom end date (ISO 8601)
tab	string	No	Report tab: overview, campaigns, photos, team, issues

Response:

```
{
  "data": {
    "storeId": "uuid",
    "dateRange": {
      "start": "2024-10-01",
      "end": "2024-12-31",
      "label": "Last 90 Days"
    },
    "kpis": {
      "complianceRate": { "value": 94, "trend": 3, "status": "green" },
      "onTimeRate": { "value": 88, "trend": 5, "status": "green" },
      "photoApprovalRate": { "value": 97, "trend": -1, "status": "green" },
      "avgCompletionDays": { "value": 4.2, "trend": -0.5, "status": "green" }
    },
    "campaignMetrics": { ... },
    "photoMetrics": { ... },
    "teamMetrics": [ ... ],
    "issueMetrics": { ... },
    "trends": [ ... ]
  },
  "meta": {
    "generatedAt": "2024-12-31T12:00:00Z",
    "comparisons": {
      "networkAverage": { ... },
      "regionAverage": { ... }
    }
  }
}
```

6.1.2 Export Store Reports

GET /api/v1/stores/{storeId}/reports/export

Query Parameters:

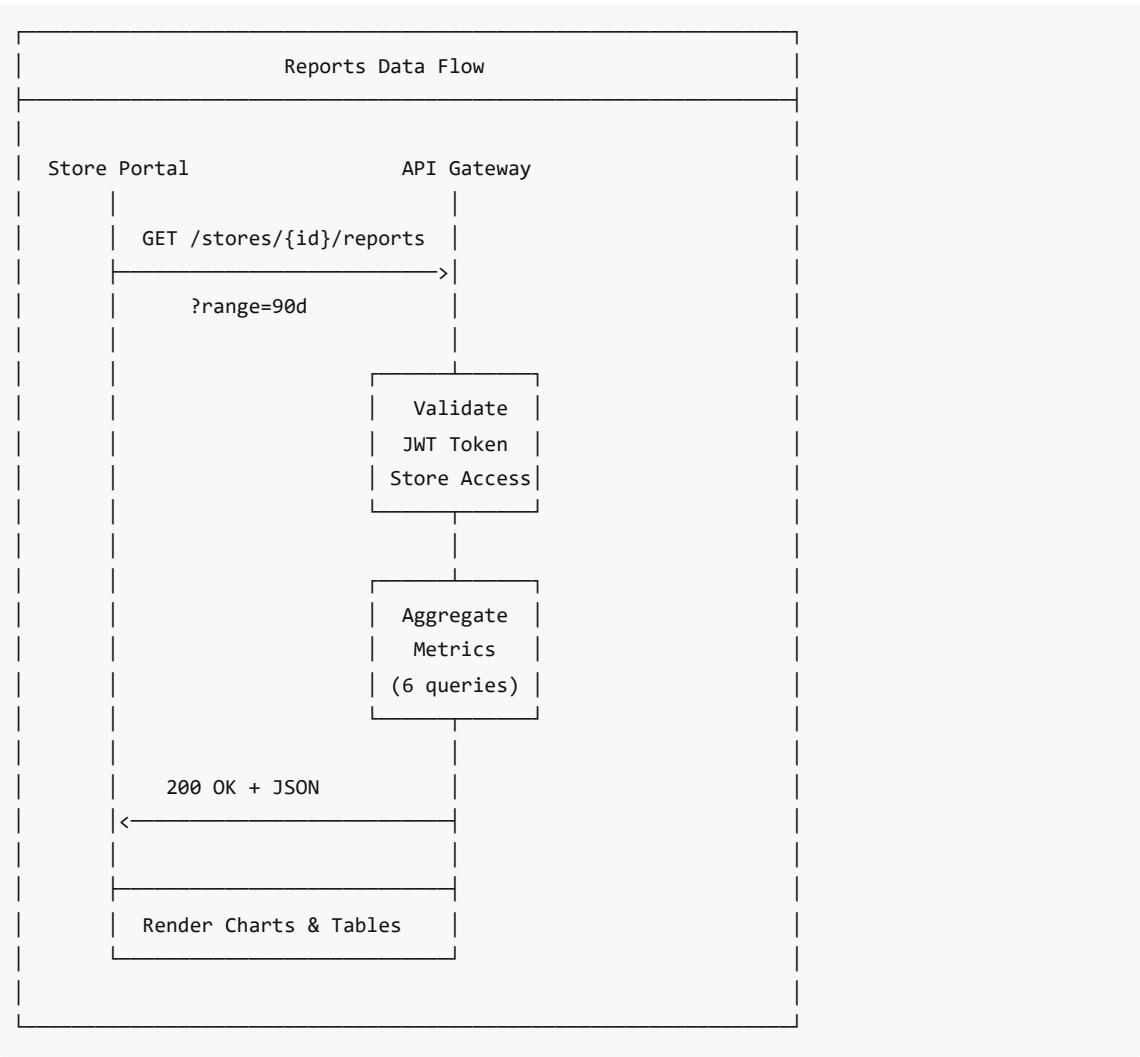
Parameter	Type	Required	Description
format	string	Yes	Export format: csv, pdf, xlsx

range	string	No	Date range (same as reports endpoint)
startDate	string	No	Custom start date
endDate	string	No	Custom end date
sections	string	No	Comma-separated sections to include

Response:

Content-Type: application/octet-stream
Content-Disposition: attachment; filename="store-reports-2024-12-31.csv"

6.2 Request/Response Flow



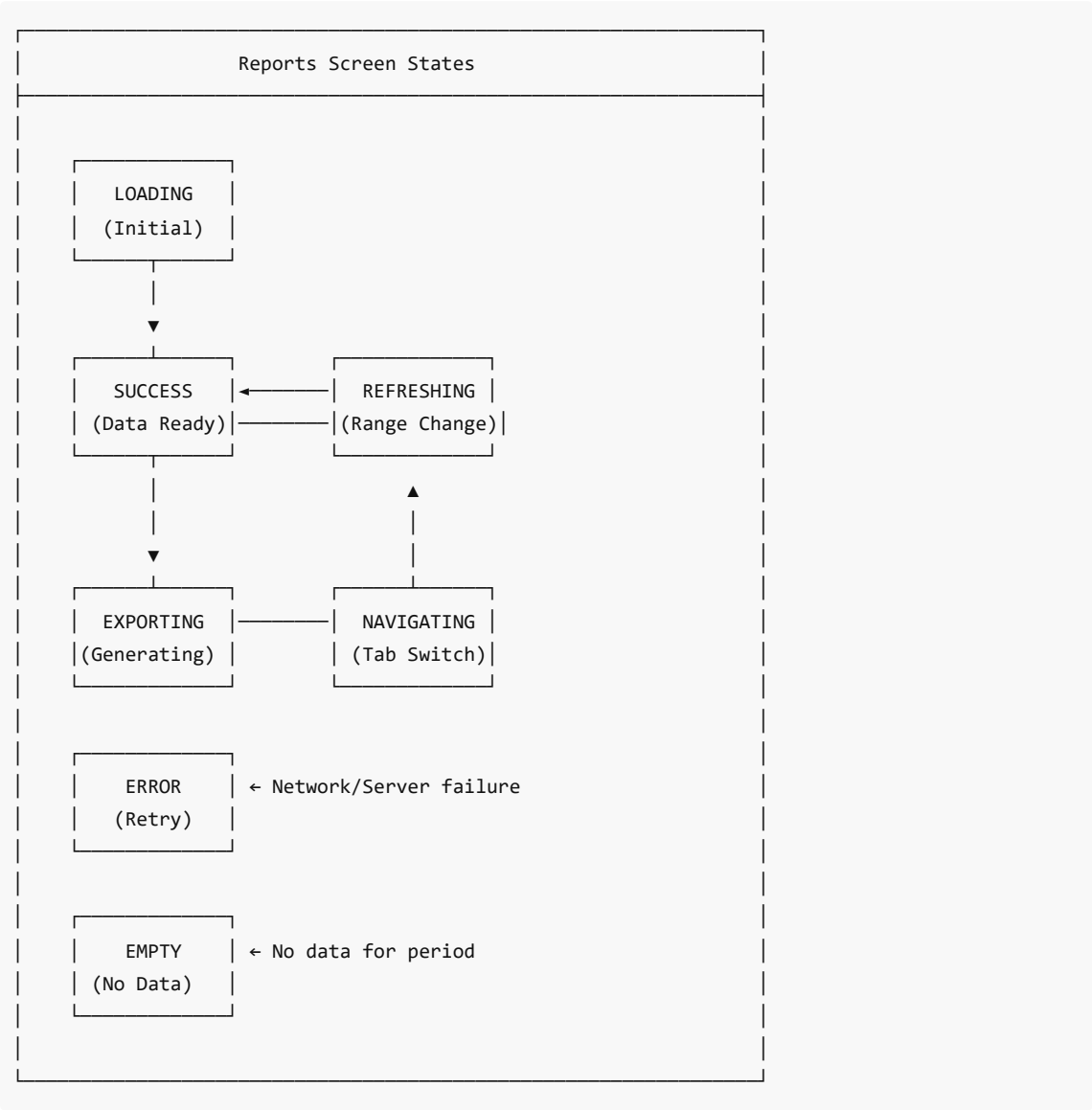
6.3 API Requirements

Requirement ID	Requirement
REQ-S005-API-001	Reports endpoint SHALL respond within 3 seconds for 90-day range

REQ-S005-API-002	Export endpoint SHALL support async generation for large date ranges
REQ-S005-API-003	API SHALL return 404 if store not found
REQ-S005-API-004	API SHALL return 403 if user lacks store access
REQ-S005-API-005	API SHALL validate date range parameters

7. State Transitions

7.1 Screen States



7.2 State Definitions

State	Description	User Actions Available
LOADING	Initial data fetch in progress	None

SUCCESS	Data loaded, charts rendered	Tab switch, date change, export
REFRESHING	Date range changed, updating	Cancel (via new action)
NAVIGATING	Switching tabs	None
EXPORTING	Report file generation	Cancel
ERROR	API failure	Retry
EMPTY	No data for selected period	Change date range

7.3 State Transition Requirements

Requirement ID	Requirement
REQ-S005-ST-001	LOADING state SHALL display skeleton loaders for all components
REQ-S005-ST-002	Tab navigation SHALL NOT trigger full page reload
REQ-S005-ST-003	Date range change SHALL trigger REFRESHING state
REQ-S005-ST-004	Export SHALL display progress indicator for large reports
REQ-S005-ST-005	ERROR state SHALL provide retry action
REQ-S005-ST-006	EMPTY state SHALL suggest expanding date range

8. Error Handling

8.1 Error Scenarios

Error Code	Scenario	User Message	Recovery Action
401	Session expired	"Your session has expired. Please log in again."	Redirect to login
403	Access denied	"You don't have permission to view reports."	Redirect to Dashboard
404	Store not found	"Store not found."	Redirect to Dashboard
422	Invalid date range	"Invalid date range. End date must be after start date."	Highlight date fields
500	Server error	"Unable to load reports. Please try again."	Show retry button
TIMEOUT	Request timeout	"Request timed out. Try a shorter date range."	Show retry button

8.2 Error Handling Requirements

Requirement ID	Requirement
REQ-S005-ERR-001	API errors SHALL display user-friendly messages

REQ-S005-ERR-002	Partial failures SHALL render available data with error notice
REQ-S005-ERR-003	Chart rendering errors SHALL display placeholder with error message
REQ-S005-ERR-004	Export failures SHALL notify user with option to retry
REQ-S005-ERR-005	Network errors SHALL be logged to error tracking service

8.3 Validation Errors

Field	Validation	Error Message
startDate	Valid ISO date	"Invalid start date format"
endDate	Valid ISO date	"Invalid end date format"
startDate	< endDate	"Start date must be before end date"
range	Max 365 days	"Date range cannot exceed 1 year"

9. Accessibility Requirements

9.1 WCAG 2.1 AA Compliance

Requirement ID	Requirement	WCAG Criterion
REQ-S005-A11Y-001	All charts SHALL have text alternatives	1.1.1 Non-text Content
REQ-S005-A11Y-002	Color SHALL NOT be only means of conveying information	1.4.1 Use of Color
REQ-S005-A11Y-003	Focus order SHALL follow logical reading order	2.4.3 Focus Order
REQ-S005-A11Y-004	Tab navigation SHALL be keyboard accessible	2.1.1 Keyboard
REQ-S005-A11Y-005	Data tables SHALL have proper header associations	1.3.1 Info and Relationships
REQ-S005-A11Y-006	Status indicators SHALL include text labels	1.3.3 Sensory Characteristics

9.2 Chart Accessibility

Requirement ID	Requirement
REQ-S005-A11Y-007	Charts SHALL provide tabular data alternative
REQ-S005-A11Y-008	Screen readers SHALL announce chart summaries
REQ-S005-A11Y-009	Color blind safe palette SHALL be used for chart colors

REQ-S005-A11Y-010	Chart patterns SHALL differentiate series beyond color
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9.3 Keyboard Navigation

Action	Keyboard Shortcut
Switch tabs	Arrow Left/Right when tabs focused
Open date picker	Enter when focused
Export report	Enter when export button focused
Navigate table rows	Arrow Up/Down
Sort table column	Enter on column header

10. Acceptance Criteria

10.1 Functional Acceptance Criteria

ID	Criterion	Verification Method
AC-S005-001	Overview tab displays 4 KPI cards with current values and trends	Manual test
AC-S005-002	Trend chart displays performance data over selected date range	Manual test
AC-S005-003	Rejection reasons pie chart displays categorized breakdown	Manual test
AC-S005-004	Team contribution bar chart shows member percentages	Manual test
AC-S005-005	Campaign table shows completion status for each campaign	Manual test
AC-S005-006	Date range selector filters all displayed data	Manual test
AC-S005-007	Export generates downloadable file in selected format	Manual test
AC-S005-008	Network/region comparison metrics display when available	Manual test
AC-S005-009	Tab navigation switches report sections without page reload	Manual test
AC-S005-010	Store Operators cannot access the Reports screen	Automated test

10.2 Non-Functional Acceptance Criteria

ID	Criterion	Verification Method
AC-S005-011	Page loads in under 3 seconds for 90-day range	Performance test
AC-S005-012	Charts render within 1 second after data load	Performance test
AC-S005-013	Export completes within 30 seconds for 365-day range	Performance test
AC-S005-014	Screen is fully accessible via keyboard navigation	Accessibility audit
AC-S005-015	Charts include text alternatives for screen readers	Accessibility audit

10.3 Edge Case Acceptance Criteria

ID	Criterion	Verification Method
AC-S005-016	New store with no data displays empty state message	Manual test
AC-S005-017	Store with single campaign displays correctly	Manual test
AC-S005-018	365-day range loads without timeout	Performance test
AC-S005-019	Charts handle zero values without rendering errors	Manual test

11. Traceability Matrix

Requirement ID	Source	Test Case	Status
REQ-S005-SEC-001	SUPP-003 RBAC	TC-S005-001	Draft
REQ-S005-UI-001	S05_Reports.md	TC-S005-002	Draft
REQ-S005-BR-001	S05_Reports.md	TC-S005-003	Draft
REQ-S005-API-001	NFR-001	TC-S005-004	Draft
REQ-S005-A11Y-001	WCAG 2.1 AA	TC-S005-005	Draft

12. Appendix

12.1 Date Range Presets

Preset	Start Date Calculation	End Date
Last 7 Days	today - 7 days	today
Last 30 Days	today - 30 days	today
Last 90 Days	today - 90 days	today
Last 365 Days	today - 365 days	today
Custom	User-selected	User-selected

12.2 Export File Formats

Format	Extension	Content
CSV	.csv	Raw tabular data, UTF-8 encoded
PDF	.pdf	Formatted report with embedded charts
Excel	.xlsx	Multi-sheet workbook with charts

12.3 Related Documents

Document	Relationship
S001_Dashboard.md	Summary metrics source
S002_Campaign_History.md	Campaign detail navigation
S003_Photo_Gallery.md	Photo detail navigation
S004_Team_Management.md	Team member source
3.1_Database_Model.md	Data model reference
4.2_Permission_Matrix.md	RBAC reference

Document Status: Draft IEEE 830 Compliance: Section 3.2 - Specific Requirements / Functional Requirements